

OCR Further Mathematics
Pure Core - Complex Numbers
Mark Scheme
AS and A Level
Further Mathematics A - H235, H245



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To be used by all students preparing for OCR AS and A Level Further Mathematics.
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Question	Answer/Indicative content	Marks	Part marks and guidance
1	Take $z = \cos\theta + i\sin\theta \Rightarrow z^{-1} = \cos\theta - i\sin\theta$ $\Rightarrow z - \frac{1}{z} = 2i\sin\theta$ $\left(z - \frac{1}{z}\right)^5 = 32i\sin^5\theta = z^5 - 5z^3 + 10z - \frac{10}{z} + \frac{5}{z^3} - \frac{1}{z^5}$ $\Rightarrow 32i\sin^5\theta = \left(z^5 - \frac{1}{z^5}\right) - 5\left(z^3 - \frac{1}{z^3}\right) + 10\left(z - \frac{1}{z}\right)$ $= 2i\sin 5\theta - 10i\sin 3\theta + 20i\sin\theta$ $\Rightarrow \sin^5\theta = \frac{5}{8}\sin\theta - \frac{5}{16}\sin 3\theta + \frac{1}{16}\sin 5\theta.$ i.e. $A = \frac{5}{8}, B = -\frac{5}{16}, C = \frac{1}{16}$ Alternative method 1 $(\sin\theta)^5 = \left(\frac{e^{i\theta} - e^{-i\theta}}{2i}\right)^5$ $= \frac{1}{(2i)^5} (e^{5i\theta} - 5e^{3i\theta} + 10e^{i\theta} - 10e^{-i\theta} + 5e^{-3i\theta} - e^{-5i\theta})$ $= \frac{1}{32i} ((e^{5i\theta} - e^{-5i\theta}) - 5(e^{3i\theta} - e^{-3i\theta}) + 10(e^{i\theta} - e^{-i\theta}))$ $= \frac{1}{16} (\sin 5\theta - 5\sin 3\theta + 10\sin\theta)$ $\Rightarrow \sin^5\theta = \frac{5}{8}\sin\theta - \frac{5}{16}\sin 3\theta + \frac{1}{16}\sin 5\theta$ $\Rightarrow A = \frac{5}{8}, B = -\frac{5}{16}, C = \frac{1}{16}$ Alternative method 2 $z = \cos\theta + i\sin\theta$ $\Rightarrow z^5 = \cos 5\theta + i\sin 5\theta$ $= (\cos\theta + i\sin\theta)^5$ $= \cos^5\theta + 5i\cos^4\theta\sin\theta - 10\cos^3\theta\sin^2\theta - 10i\cos^2\theta\sin^3\theta + 5\cos\theta\sin^4\theta + i\sin^5\theta$ $\Rightarrow \sin 5\theta = 5\cos^4\theta\sin\theta - 10\cos^2\theta\sin^3\theta + \sin^5\theta$ $= 5(1 - \sin^2\theta)^2\sin\theta - 10(1 - \sin^2\theta)\sin^3\theta + \sin^5\theta$ $= 5\sin\theta - 10\sin^3\theta + 5\sin^5\theta - 10\sin^3\theta + 10\sin^5\theta + \sin^5\theta$ $= 5\sin\theta - 20\sin^3\theta + 16\sin^5\theta$ $z^3 = \cos 3\theta + i\sin 3\theta$ $= (\cos\theta + i\sin\theta)^5 = \cos^3\theta +$	M1 (AO 2.1) A1 (AO 1.1) M1 (AO 1.1) A1 (AO 2.2a) Alternative method 1 M1 A1 M1 A1 Alternative method 2 M1 A1 M1 A1 [4]	Use of z and de Moivre Sight of i is necessary Both sides Attempt conversion into $\sin$ soi All three stated Alternative method 1 Sight of i in the denominator is necessary Collection to convert back All three stated Alternative method 2 De Moivre for both Eliminate $\sin^3\theta$ All three stated



Question			Answer/Indicative content	Marks	Part marks and guidance	
			$3i \cos^2 \theta \sin \theta - 3 \cos \theta \sin^2 \theta - i \sin^3 \theta$ $\Rightarrow \sin 3\theta = 3 \cos^2 \theta \sin \theta - \sin^3 \theta$ $= 3 \sin \theta - 4 \sin^3 \theta$ $\Rightarrow \sin 5\theta - 5 \sin 3\theta = -10 \sin \theta + 16 \sin^5 \theta$ $\Rightarrow 16 \sin^5 \theta = 10 \sin \theta - 5 \sin 3\theta + \sin^5 \theta$ $\Rightarrow A = \frac{10}{16} = \frac{5}{8}, B = -\frac{5}{8}, C = \frac{1}{16}$			
			Total	4		
2			e.g. $\alpha^2 + \beta^2 + \gamma^2 = -5$ means that at least one root is complex But complex roots come in complex pairs so there are 2 complex roots. Given that there are 3 roots and 2 are complex one is real.	B1 (AO 2.4) B1 (AO 2.4) B1 (AO 2.4) [3]		
			Total	3		



Question			Answer/Indicative content	Marks	Part marks and guidance	
3	a	i	Circle Centre $1-2i$, Radius 3	B1 (AO 1.1) B1 (2.2a) [2]	Be generous over circles drawn freehand If the axes are scaled then a mark at $(1, -2)$ will do. For radius, an indication that the radius is 3 will do (e.g. passing through $(4, -2)$ etc if marked will do.)	
		ii	Straight vertical line $x = \frac{1}{2}$	B1 (AO 1.1) B1 (2.2a) [2]	Can be seen by $x = \frac{1}{2}$ being labelled on the axis and vertical line through it	
	b		Inside circle And to the left of $x = \frac{1}{2}$	B1 (AO 1.1) B1 (2.2a) [2]	Or their line if it is vertical.	
			Total	6		

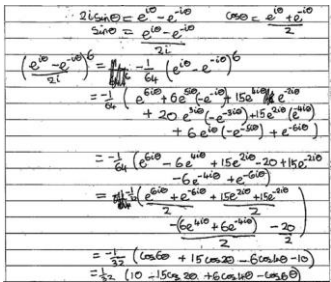


Question			Answer/Indicative content	Marks	Part marks and guidance	
4	a	i	DR $3z_1 + 4z_2 = 3(3 - 7i) + 4(2 + 4i) = 17 - 5i$	B1 (AO1.1) [1]		
		ii	DR $z_1 z_2 = (3 - 7i)(2 + 4i) = 6 + 12i - 14i - 28(-1)$ $= 34 - 2i$	M1 (AO1.1) A1 (AO1.1) [2]	Attempted expansion with $i^2 = -1$ used and at least 3 correctly expanded terms	- 28(-1) can be simply +28
		iii	DR $\frac{z_1}{z_2} = \frac{3-7i}{2+4i} = \frac{3-7i}{2+4i} \times \frac{2-4i}{2-4i}$ $= \frac{6-12i-14i-28}{4+16} = \frac{-22-26i}{20} = -\frac{11}{10} - \frac{13}{10}i$	M1 (AO1.1) A1 (AO1.1) [2]	Multiplying top and bottom by (real multiple of) conjugate of bottom. Must see some evidence of expansion	All $\frac{11-13i}{10}$ or $-\frac{11+13i}{10}$ ow
	b		DR $\sqrt{3^2 + (-7)^2}$ or $\tan^{-1}\left(\frac{-7}{3}\right)$ $ z_1 = \sqrt{58}$ or awrt 7.62 or $\arg z_1 =$ awrt -1.17 or 5.12 rads $z_1 = \sqrt{58}\text{cis}(-1.17)$ or $z_1 = \sqrt{58}e^{-1.17i}$ or $z_1 = \sqrt{58}(\cos(-1.17) + i\sin(-1.17))$ or $[\sqrt{58}, -1.17]$	M1 (AO1.1) A1 (AO1.1) A1 (AO2.5) [3]	Explicit working must be seen Must be in correct form with $\sqrt{58}$ exact and could be awrt 5.12 instead of -1.17.	Other trig calculations could be sufficient for M1 provided that these are being used to find the argument. Do not condone degrees Condone round brackets
			Total	8		



Question		Answer/Indicative content	Marks	Part marks and guidance		
5	a	DR $\sin \theta = \frac{e^{i\theta} - e^{-i\theta}}{2i}$ $\sin^6 \theta = \left(\frac{e^{i\theta} - e^{-i\theta}}{2i} \right)^6 = -\frac{1}{64} (e^{i\theta} - e^{-i\theta})^6$ $(e^{i\theta} - e^{-i\theta})^6 =$ $e^{6i\theta} - 6e^{4i\theta} + 15e^{2i\theta} - 20 + 15e^{-2i\theta} - 6e^{-4i\theta} + e^{-6i\theta}$ $e^{6i\theta} + e^{-6i\theta} - 6(e^{4i\theta} + e^{-4i\theta}) + 15(e^{2i\theta} + e^{-2i\theta}) - 20$ $= 2 \cos 6\theta - 6 \times 2 \cos 4\theta + 15 \times 2 \cos 2\theta - 20$ $\therefore \sin^6 \theta =$ $-\frac{1}{64} (2 \cos 6\theta - 12 \cos 4\theta + 30 \cos 2\theta - 20)$ $= \frac{1}{32} (10 - 15 \cos 2\theta + 6 \cos 4\theta - \cos 6\theta)$	*B1(AO1.1a)M1(AO2.1)M1(AO1.1)M1(AO2.1)dep*A1(AO1.1)[5]	Raising expression for $\sin \theta$ to the power 6 and $(2i)^6 = -64$. Genuine attempt to use binomial expansion with correct evaluated binomial coefficients. Condone sign errors Collecting terms and using $e^{i\phi} + e^{-i\phi} = 2 \cos \phi$ at least once. AG. Fully correct argument <u>Examiner's Comments</u> A considerable number of candidates derived this correctly, although sometimes losing a mark through omitting sufficient detailed reasoning. This usually happened where candidates did not explicitly group corresponding positive and negative exponentials together before converting to the cosine forms. One	Condone $2i$ $\sin \theta = \frac{e^{i\theta} - e^{-i\theta}}{2i}$ Allow use $\sin \theta = \frac{e^{i\theta} + e^{-i\theta}}{2i}$ For 1st two M marks only If i omitted from denominator their expression for $\sin \theta$ then only this M mark can still be awarded	



Question	Answer/Indicative content	Marks	Part marks and guidance
			surprisingly common error was to quote the complex exponential expression for $\sin\theta$ with 2 in the denominator, rather than $2i$. This is such a fundamental mistake that it cost candidates, but they could go on to pick up a method mark were they to then binomially expand $(e^{i\theta} - e^{-i\theta})^6$. The standard abbreviation 'z' was often used in place of $e^{i\theta}$, which was acceptable and efficient if defined as such, but penalised otherwise. A small number of candidates dispensed with the binomial theorem and attempted to expand by repeated multiplication, usually incorrectly. The binomial expansion, along with other formulae in the formula booklet, can be used without justification even when detailed reasoning is requested. Where candidates expanded $(\cos\theta + i\sin\theta)^6$, they were unable to make progress deemed worthy of marks. Exemplar 3  In this example we see what is otherwise a clear argument marred by the absence of a key element –



Question			Answer/Indicative content	Marks	Part marks and guidance	
					namely that they never equate $\sin^6\theta$ to their solution. Demonstrations and proofs must explicitly show that what is posited has been proved.	
	b		DR $\theta = \frac{\pi}{8}$ and eg $\cos 2\theta = \frac{\sqrt{2}}{2}$ $\sin^6 \frac{\pi}{8} = \frac{1}{32} \left(10 - 15 \times \frac{\sqrt{2}}{2} - \frac{-\sqrt{2}}{2} (+6(0)) \right)$ $\sin \frac{\pi}{8} = \sqrt[6]{\frac{1}{64} (20 - 15\sqrt{2} + \sqrt{2})}$ $= \frac{1}{2} \sqrt[6]{20 - 14\sqrt{2}}$	*M1(AO2.1)dep*M1(AO1.1)A1(AO2.2a)[3]	Choice of θ soi and calculation of at least one cos term. Substitution and calculation of all cos terms AG Some intermediate working must be seen	Terms must be shown distinct either in this line or in the form of $\cos n \frac{\pi}{8}$

Examiner's Comments

This question was well answered, with most

candidates substituting

$\theta = \frac{\pi}{8}$ into the formula and explicitly evaluating the cosines, although sometimes with a sign error in the $\cos 6\theta$ term. This being a detailed reasoning question, candidates had to show sufficient working to convince that they had indeed evaluated the separate terms and not simply worked back from the given answer. Likewise, some intermediate step, such as writing $\frac{1}{32}$ as



Question			Answer/Indicative content	Marks	Part marks and guidance	
					$\frac{2}{64}$, was required to justify the factor of $\frac{1}{2}$ in the final answer.	
			Total	8		



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Question		Answer/Indicative content	Marks	Part marks and guidance		
6	a	$(2 + 3i) - (1 - i) (= \pm(1 + 4i))$ so $ 1 + 4i = \sqrt{1^2 + 4^2}$ 17	B1(AO1.1) M1(AO2.2a) A1(AO1.1) [3]	Eitherway round Can be implied by vector Or finding the square of their side		
<u>Examiner's Comments</u> This part was usually done correctly. Most candidates subtracted the two numbers, occasionally with errors and used the correct method to find the distance. Complex number notation was often replaced by vector notation and / or a diagram. The great majority of candidates then squared their distance to find the area, although a small number thought that the given points were opposite, rather than adjacent, and hence arrived at half the correct answer.						

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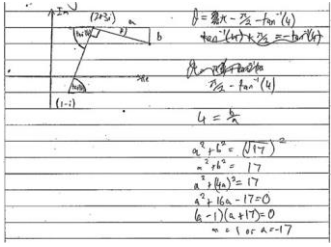
Question		Answer/Indicative content	Marks	Part marks and guidance		
	b	$(\pm i) \times (\pm(1 + 4i))$ $(2 + 3i) \pm i(1 + 4i)$ and $(1 - i) \pm i(1 + 4i)$ So vertices at -3 and $-2 + 4i$ Or at $5 - 2i$ and $6 + 2i$	*M1(AO3.1a)dep* M1(AO2.2a)A1(AO3.2a)A1(AO3.2a) [4]	Method to find a complex number representing perpendicular side. Can be implied by $\pm(4 - i)$ Method to find both pairs of numbers Both clearly paired and in complex number form for final A1	Or vector form if take geometric approach If M1M0A0A0 then add SC1 for any two correct vertices	

Examiner's Comments

There were many correct answers to this part, although most candidates proceeded geometrically (either in vector notation or diagrammatically) to find the perpendicular vector / s, rather than multiplying $1 + 4i$ by $(\pm)i$. Many candidates then translated and found the correct vertices, although some did leave them in vector notation, rather than giving the (complex) numbers. The majority, but by no means all, showed the correct pairings for the numbers. A few candidates attempted the circuitous route of finding the Cartesian equation of one or both of the perpendicular sides and considering the points on

such at distance $\sqrt{17}$ from the given points. This rarely met with success, with



Question			Answer/Indicative content	Marks	Part marks and guidance
					candidates either giving up or becoming embroiled in miscalculations. Exemplar 2  Here is an example of a candidate who tries to find the translation vector via simultaneous equations using gradient and distance. They get the gradient wrong and neglect to fully square (4a) resulting in the breakdown of the method. Selecting efficient methods is one key to success in these papers.
			Total	7	

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Question		Answer/Indicative content	Marks	Part marks and guidance		
7	a	DR $r^2 = (-4)^2 + (\sqrt{48})^2$ or $(r \cos \theta = -4$ and $r \sin \theta = \sqrt{48})$ or $\tan \theta = -\sqrt{3}$ oe $r = 8$ (ie $z = 8e^{i\theta}$) $\theta = 2\pi/3$ (ie $z = re^{i2\pi/3}$) $\sqrt[3]{8}$ or 2 $\frac{2\pi}{9}$ soi $\frac{2\pi}{3} + 2\pi k$ for $k = 1$ and 2 oe seen $2e^{\frac{2}{9}\pi i}$, $2e^{\frac{8}{9}\pi i}$ and $2e^{-\frac{4}{9}\pi i}$	M1 (AO2.1) A1 (AO1.1) N1ft (AO2.1) B1ft (AO2.1) M1 (AO2.2a) A1 (AO1.1) [6]	Correct use of relevant formula(e). Some working must be seen. Not ± 8 unless later corrected Modulus of cube root(s) is the cube root of their modulus Argument of (principal) cube root is one third of their argument Considering further arguments at angular distance 2π o $2e^{\frac{1}{9}\pi i}$ a $2e^{\frac{14}{9}\pi i}$ r n d	Correct answer with no working: M0A0 or eg $\theta = 8\pi/3$ Must be in exponential form, not just $r =$ and $\theta =$. Do not condone any missing i's.	
	b	DR The cube roots form an equilateral triangle which has (3) lines of symmetry, (one) through each vertex $\theta = \frac{2\pi}{9}$, $\theta = \frac{8\pi}{9}$ and $\theta = -\frac{4\pi}{9}$ soi	B1 (AO2.2a) B1 (AO2.2a) [3]	for one for all three without extras	ft their angles if $2\pi/3$ apart. If valid alternatives, must come from clear explanation / diagram	



Question			Answer/Indicative content	Marks	Part marks and guidance
			Total	9	



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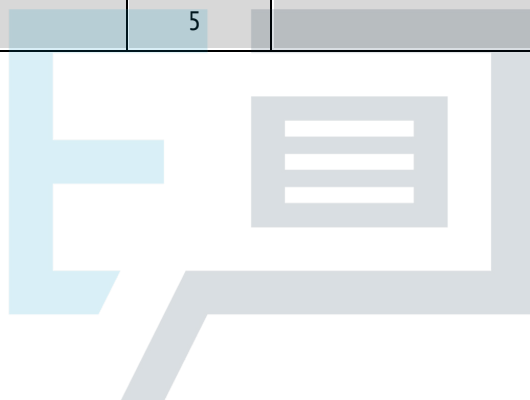
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Question			Answer/Indicative content	Marks	Part marks and guidance	
8			DR $z = \frac{-20 \pm \sqrt{(-20)^2 - 4 \times 4 \times 169}}{2 \times 4}$	M1 (AO1.1)	Term by term substituting into formula.	Condone anything correct of the for $\frac{p \pm \sqrt{q}}{r}$
			$z = \frac{5 \pm 12i}{2}$ $r = \sqrt{\left(\frac{5}{2}\right)^2 + \left(\frac{12}{2}\right)^2} = \frac{13}{2}$ $\theta = \tan^{-1} \frac{6}{2.5}$	A1 (AO1.1) B1ft (AO1.1)	If formula quoted, allow one slip ... Or correctly completes the square	$e^{4\left(\left(z - \frac{5}{2}\right)^2 - \frac{25}{4}\right) + 169 - 0}$ g
			$\frac{13}{2}(\cos 1.18 + i \sin 1.18)$	M1 (AO1.1)		Ft workings from complex conjugate distinct pair (with real component)
			$\frac{13}{2}(\cos(-1.18) + i \sin(-1.18))$	A1 (AO2.5) [5]	Attempting to find argument using trigonometry Angle must be in radians.	$\theta = \cos^{-1} \frac{2.5}{6.5}, \theta = \sin^{-1} \frac{6}{6.5}$ oe eg $\frac{13}{2} \text{cis} 1.18$ or $\left[\frac{13}{2}, 1.18\right]$
					Argument could be 5.11 but both angles must be the same.	Not 5.10 (rounding error) Not e.g. $\cos(-1.18) + i \sin(5.11)$



Question			Answer/Indicative content	Marks	Part marks and guidance	
					<u>Examiner's Comments</u> The detailed reasoning command in this question means that algebraic methods must be clearly shown. It must be evident that the form $a + bi$ is not merely extracted from a calculator equation solver and that modulus-argument form is not merely extracted from calculator complex number conversion. Note that $\cos\theta - i\sin\theta$ is not an acceptable modulus-argument form.	
			Total	5		



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Question			Answer/Indicative content	Marks	Part marks and guidance	
9	a		$\beta = 1 - i\sqrt{2}$ oe	B1 (AO2.2a) [1]		
	b		$\alpha\beta\gamma = \frac{3}{2}, \alpha\beta = (1+i\sqrt{2})(1-i\sqrt{2}) = 3$ $\Rightarrow \gamma = \frac{1}{2}$	M1 (AO2.1) A1 (AO1.1) [2]	Use of $\alpha\beta\gamma = \frac{3}{2}$ to find the 3rd root. Alternatively, find $x^2 - 2x + 3$ and divide	
	c		$(x - (1+i\sqrt{2}))(x - (1-i\sqrt{2}))(2x-1) = 0$ $\Rightarrow (x^2 - 2x + 3)(2x - 1) = 0$ $\Rightarrow 2x^3 - 4x^2 + 6x - x^2 + 2x - 3 = 0$ $\Rightarrow 2x^3 - 5x^2 + 8x - 3 = 0$ i.e. $p = -5, q = 8$ Alternate Method $\alpha + \beta + \gamma = -\frac{a}{2} = \frac{5}{2} \Rightarrow p = -5$ $\alpha\beta + \beta\gamma + \gamma\alpha = \frac{q}{2}$ $= \frac{1}{2}(1+i\sqrt{2}+1-i\sqrt{2}) + (1+i\sqrt{2})(1-i\sqrt{2})$ $= 1 + 3 = 4 \Rightarrow q = 8$	M1 (AO3.1a) A1 (AO1.1) M1 A1 [2]	Multiply out (can be seen in (b)) Use of symmetry forms for roots	

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Question		Answer/Indicative content	Marks	Part marks and guidance	
	d	$\alpha = 1 + i\sqrt{2} = \sqrt{3} \left(\frac{1}{\sqrt{3}} + i \frac{\sqrt{2}}{\sqrt{3}} \right) = 3^{\frac{1}{2}} (\cos \theta + i \sin \theta)$ where $\cos \theta = \frac{1}{\sqrt{3}}$ $\sin \theta = \frac{\sqrt{2}}{\sqrt{3}} \Rightarrow \tan \theta = \sqrt{2}$ re $\beta = 1 - i\sqrt{2} = \sqrt{3} \left(\frac{1}{\sqrt{3}} - i \frac{\sqrt{2}}{\sqrt{3}} \right) = 3^{\frac{1}{2}} (\cos \theta - i \sin \theta)$ oe $\Rightarrow \alpha^n = 3^{\frac{n}{2}} (\cos n\theta + i \sin n\theta)$, $\beta^n = 3^{\frac{n}{2}} (\cos n\theta - i \sin n\theta)$ $\Rightarrow \alpha^n + \beta^n = 2 \times 3^{\frac{n}{2}} \times \cos n\theta$ AG $\Rightarrow \alpha^n + \beta^n + \gamma^n = 2 \times 3^{\frac{n}{2}} \times \cos n\theta + \frac{1}{2^n}$	M1 (AO2.1) A1 (AO1.1) M1 (AO2.1) A1 (AO2.1) [4]	Either α or β seen in mod / arg form For both of them - accept exponentials Derivation of α^n or β^n <u>Examiner's Comments</u> In part (b), rather than using the symmetry of roots, the working for part (c) was seen. So, some extra algebra was needed, creating more opportunities for errors. Part (d) proved to be a challenging question for most candidates. It was necessary to realise that the only way to find a simple answer for the nth power of each complex root was by a conversion to mod / arg form. Those that did realise this usually succeeded in obtaining the required response.	
		Total	9		

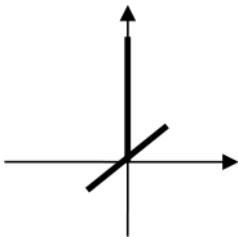


Question			Answer/Indicative content	Marks	Part marks and guidance	
10			$\left(z^2 + \frac{1}{z^2}\right)^3 = z^6 + 3z^2 + \frac{3}{z^2} + \frac{1}{z^6}$ $z^2 + \frac{1}{z^2} = 2 \cos 2\theta \Rightarrow \left(z^2 + \frac{1}{z^2}\right)^3 = 8 \cos^3 2\theta$ an $z^2 + \frac{1}{z^2} = 2 \cos 2\theta$ an $z^6 + \frac{1}{z^6} = 2 \cos 6\theta$ d d $\text{an } \left(z^2 + \frac{1}{z^2}\right)^3 = \left(z^6 + \frac{1}{z^6}\right) + 3\left(z^2 + \frac{1}{z^2}\right)$ d $\Rightarrow 8 \cos^3 2\theta = 2 \cos 6\theta + 6 \cos 2\theta$ $\Rightarrow 4 \cos^3 2\theta = \cos 6\theta + 3 \cos 2\theta$ AG	M1 (AO1.1) A1 (AO1.1) M1 (AO3.1a) A1 (AO2.1) A1 (AO1.1) [5]	Use of binomial use of De Moivre for any one term <u>Examiner's Comments</u> Most candidates showed that they knew, and could correctly use, the identity $\cos n\theta = \frac{1}{2}(z^n + z^{-n}).$ However, a significant number of worked out $(z^2 + z^{-2})^2$ before finding the four terms of $(z^2 + z^{-2})^3$, with errors often seen in the algebra. In 'show that' questions, where the answer is given, then marks are awarded for the accuracy of the complete mathematical argument.	
			Total	5		



Question		Answer/Indicative content	Marks	Part marks and guidance	
11	a	$25i = 25e^{2i}$ $25i = 25e^{\frac{\pi}{2}i}$ $\Rightarrow \sqrt{25i} = 5e^{\frac{\pi}{4}i}$ an $5e^{\frac{5\pi}{4}i}$ d So the square $5e^{\frac{\pi}{4}i}$ an $5e^{\frac{5\pi}{4}i}$ roots are d Alternate Method $\sqrt{25i} = a + bi$ $\Rightarrow a = b = \pm \frac{5}{2}\sqrt{2}$ $\Rightarrow \sqrt{25i} = \pm \frac{5}{2}\sqrt{2}(1+i) = \pm 5\left(\cos\frac{1}{4}\pi + i\sin\frac{1}{4}\pi\right)$ $\Rightarrow \sqrt{25i} = 5e^{\frac{1}{4}\pi}$ an $5e^{\frac{5}{4}\pi}$ d	B1 (AO3.3) M1 (AO3.1a) A1 (AO3.2a) B1 M1 A1 [3]	Conversion soi Attempt to find at least one root by rooting the modulus and halving the argument soi Both Conversion and attempt to square a and b. Ignore $\pm$ Both	

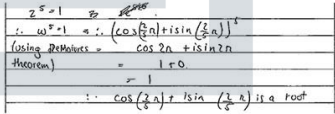


Question			Answer/Indicative content	Marks	Part marks and guidance
	b			B1 (AO1.1) [1]	All three but no extras. Scales etc are not required but if no scale then the lines representing the roots should be at 45° to axis. Accept points. No extras Line representing $25i$ must be at least two times as long <u>Examiner's Comments</u> In part (a) most candidates were able to write $25i$ in exponential form but found difficulty in finding the square root. The first root was found easily but unlike real numbers the addition of $\pm$ in front of the first root does not give the correct response. A few students did not give their answer for θ in the range stated in the question, $0 \leq \theta < 2\pi$, writing the argument $\text{as } -\frac{3\pi}{4}.$ The Argand diagram in part (b) was poorly done. Not only were the two roots expected in the right place but that the modulus of $25i$ on the y-axis needed to be considerably larger than the modulus of each root.
			Total	4	

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Question		Answer/Indicative content	Marks	Part marks and guidance	
12	a	DR $\omega = \cos \frac{2\pi}{5} + i \sin \frac{2\pi}{5}$ $\Rightarrow \omega^5 = \left(\cos \frac{2\pi}{5} + i \sin \frac{2\pi}{5} \right)^5$ $= \cos 2\pi + i \sin 2\pi = 1 + 0i$ $= 1$	M1(AO2.1) A1(AO1.1) [2]	Use of exponential s is satisfactory Could be argued backwards Finding ω^5 AG <u>Examiner's Comments</u> This part, as with the other parts of this question, saw many candidates effectively write down the answer and then to say 'so this is the answer'. So, in this part, to write 'ω is a root of the equation so ω is a root' is not credit worthy. It was expected that the de Moivre's theorem would be used. <u>Exemplar 2</u>  This candidate substituted ω for z and showed that $\omega^5 = 1$.	
	b	$\omega^2, \omega^3, \omega^4, 1$	B1(AO1.1) [1]	Alternative: Roots are $\cos \frac{2k\pi}{5} + i \sin \frac{2k\pi}{5}$ for $k = 2, 3, 4$ and 1 (or $k = 5$) <u>Examiner's Comments</u> Answers were accepted in various forms including arguments in the range $-\pi \leq \theta \leq \pi$.	Exponential s satisfactory



Question		Answer/Indicative content	Marks	Part marks and guidance		
	c	DR $\omega^5 - 1 = 0$ $\Rightarrow (\omega - 1)(\omega^4 + \omega^3 + \omega^2 + \omega + 1) = 0$ $\Rightarrow \omega^4 + \omega^3 + \omega^2 + \omega = -1$ Alternatively: $1 + \omega + \omega^2 + \omega^3 + \omega^4 =$ $\frac{1 - \omega^5}{1 - \omega} = \frac{0}{1 - \omega} \text{ M1}$ or $\omega + \omega^2 + \omega^3 + \omega^4$ $\omega \left(\frac{1 - \omega^4}{1 - \omega} \right) = \left(\frac{\omega - \omega^5}{1 - \omega} \right) = \left(\frac{\omega - 1}{1 - \omega} \right) = -1$ Alternatively: sum of roots $= -\frac{b}{a}$ where $b = 0$ M1- needs explanation - i.e. coefficient of z^4 term $= 0$	M1(AO1. 1a) A1(AO2. 1) [2]	Use equation and ω AG <u>Examiner's Comments</u> A response of '$\omega + \omega^2 + \omega^3 + \omega^4 + 1 = 0$, therefore $\omega + \omega^2 + \omega^3 + \omega^4 = -1$' was not acceptable. Candidates who were successful in this part either used the formulae for a geometric progression or argued that the sum of the roots of an equation equals $-\frac{b}{a}$ where b is the coefficient of the z^4 term, which is 0. A common response was to use the calculator, thus bypassing the working required to be seen in a 'show that...' question.		



Question		Answer/Indicative content	Marks	Part marks and guidance		
	d	AG $\left(\omega + \frac{1}{\omega}\right)^2 + \left(\omega + \frac{1}{\omega}\right) - 1 = \omega^2 + 2 + \frac{1}{\omega^2} + \omega + \frac{1}{\omega} - 1$ $= \frac{1}{\omega^2}(\omega^4 + \omega^2 + 1 + \omega^3 + \omega) = 0$ Since $\frac{1}{\omega^2} \neq 0$, $\omega^4 + \omega^2 + 1 + \omega^3 + \omega = 0$ or from part (c) Alternatively: $\omega^4 + \omega^3 + \omega^2 + \omega + 1 = 0$ $\Rightarrow \omega^2 \left(\omega^2 + \omega + 1 + \frac{1}{\omega} + \frac{1}{\omega^2} \right) = 0$ M1 $\Rightarrow \omega^2 \left(\left(\omega^2 + 2 + \frac{1}{\omega^2} \right) + \left(\omega + \frac{1}{\omega} \right) + 1 - 2 \right) = 0$ A1 $\Rightarrow \omega^2 \left(\left(\omega + \frac{1}{\omega} \right)^2 + \left(\omega + \frac{1}{\omega} \right) - 1 \right) = 0$ $\left(\omega + \frac{1}{\omega} \right)^2 + \left(\omega + \frac{1}{\omega} \right) - 1 = 0$ A1 Since $\omega^2 \neq 0$,	M1(AO2.1) A1(AO1.1) A1(AO2.2a) [3]	Multiply out For extraction of ω^2 For dealing with the 2 <u>Examiner's Comments</u> It was unfortunate to see a number of candidates writing down the left hand expression = 0 through their working. Additionally, candidates would multiply (or divide) by ω^2 without commenting on the value of the function. As in the previous parts, writing down values obtained from a calculator was not acceptable.		



Question		Answer/Indicative content	Marks	Part marks and guidance		
	e	$\frac{1}{\omega} = \cos \frac{2\pi}{5} - i \sin \frac{2\pi}{5}$ $\Rightarrow \left(\omega + \frac{1}{\omega} \right) = 2 \cos \frac{2\pi}{5}$ From (iii) solving quadratic: $\left(\omega + \frac{1}{\omega} \right) = \frac{-1 \pm \sqrt{5}}{2}$ $\Rightarrow 2 \cos \frac{2\pi}{5} = \frac{\sqrt{5}-1}{2} \Rightarrow \cos \frac{2\pi}{5} = \frac{\sqrt{5}-1}{4}$ $= -\frac{1}{4} + \frac{\sqrt{5}}{4} \text{ or}$ $-\frac{1}{4} + \frac{1}{4}\sqrt{5} \text{ or}$ $-0.25 + 0.25\sqrt{5}$	B1(AO3.1a) B1(AO3.1a) M1(AO2.2a) A1(AO2.3) [4]	$\omega + \frac{1}{\omega}$ may be seen in (d) BC Equating For taking the valid value and presenting in correct form oe No other forms acceptable <u>Examiner's Comments</u> Most candidates were able to convert the quadratic equation in part (d) to one involving $2\cos \frac{2\pi}{5}$ to obtain the correct value, although a few did not recognise that one of the roots of the equation was not valid and should not have been quoted as part of the solution.		
		Total	12			



Question	Answer/Indicative content	Marks	Part marks and guidance	
13	DR Second root is the conjugate of $2 + 5i$, so $x = 2 - 5i$ so So the cubic $x^3 - 2x^2 + 21x + 58 = 0$ can be written $(x - a)(x - (2 + 5i))(x - (2 - 5i)) = 0$ $\Rightarrow -a(2 + 5i)(2 - 5i) = 58$ $\Rightarrow a = -2$ So the solution of the cubic is $\Rightarrow x = -2, 2 \pm 5i$	B1(AO2. 2a) M1(AO1. 1) A1(AO2. 1) A1(AO1. 1) [4]	Soi by correct quadratic $x^2 - 4x + 29$ Attempt to factorise using complex conjugate Shown convincingly or (i.e. $(x + 2)$ seen <u>Examiner's Comments</u> Candidates stated correctly that the complex conjugate of $2 + 5i$ (i.e. $2 - 5i$) was also a root. The process of finding the third root was varied. The most elegant (and requiring the least work) was to state that the product of all three roots is -58 meaning that the third root has to be $x = -2$. The longer (but more popular) way was to create the quadratic function with the two given factors and to divide that into the cubic function to obtain $x + 2$ and hence the third root. A number of candidates did not answer the question asked, leaving their answer	Any valid method to find real root by reasoning, including division, or listing or using the factor theorem or sum of roots. NB. A DR question so full reasoning must be shown.



Question			Answer/Indicative content	Marks	Part marks and guidance	
					in terms of factors of the original cubic function.	
			Total	4		

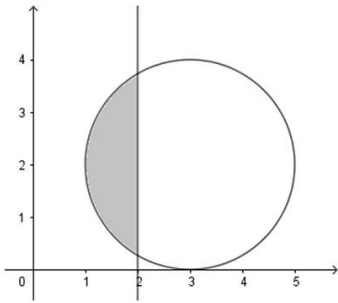


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Question		Answer/Indicative content	Marks	Part marks and guidance		
14			B1(AO2.2a) B1(AO2.2a) B1(AO1.1) [3]	Correct circle Centre $3 + 2i$ radius 2 Correct line $\text{Re}(z) = 2$ Correct shading.	“correct” requires clear indication of centre and touching x-axis Alternative y, candidates may shade the region that is not required, but should indicate clearly that what they have shaded is not required.	

Examiner's Comments

Candidates should understand that while only a sketch is required for this question (they are not required to bring into the examination a pair of compasses, for instance), it needs to be clear to examiners that what has been sketched really is a circle and a straight line. A few candidates lost one or more marks because of the lack of clarity here.

A number of candidates treated the inequality $|z| \leq |z - 4|$ as if it were $x \leq 4$ and so shaded the incorrect region.



Question			Answer/Indicative content	Marks	Part marks and guidance	
					In this question it was acceptable to shade either the region that was required or the region that was not required.	
			Total	3		
15		a	DR $\arg(z) = \frac{\pi}{6}$ $\Rightarrow \arg(z^n) = \frac{n\pi}{6} = 2\pi$ $\Rightarrow n = 12$	B1 (AO 1.1) M1 (AO 2.1) A1 (AO 1.1) [3]	Equating $\arg$ to 2π	
		b	4096 or 2^{12}	B1 (AO 1.1) [1]		
			Total	4		



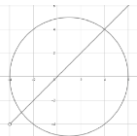
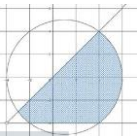
Question			Answer/Indicative content	Marks	Part marks and guidance	
16		a	The x - z plane	B1 (AO 2.2a) [1]	or $y = 0$	
		b	$\frac{2a - a^2}{3} = -1$ $a^2 - 2a - 3 = 0 \Rightarrow a = -1, 3$ $a > 0 \Rightarrow a = 3$	B1 (AO 1.1) M1 (AO 3.1a) A1 (AO 2.3) [3]	BC. Rearranging the quadratic equation and solving. discarding $a = -1$	
		c	Any reflection is self-inverse... oe ...so $A^2 = AA^{-1} = I$	B1 (AO 2.4) B1 (AO 2.4) [2]	eg "If you do a reflection twice it gets back to where it started"	
			Total	6		

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Question			Answer/Indicative content	Marks	Part marks and guidance		
17		a	C₁: Circle, centre (1,0) radius 5 C₂: Half line starting at (but not including) (-4, -4) Gradient 1 (i.e. through (0, 0))	B1 (AO1.2) B1 (AO1.1) B1 (AO2.2a) B1 (AO1.1) [4]	Soi eg by a circle through points (6,0) and (-4, 0)		
		b	Shading: Inside circle Under line but above horizontal through (-4, -4). But nothing extra	B1 (AO2.2a) B1 (AO2.2a) [2]	Withhold one mark for anything extra or omissions		
			Total	6			



Question			Answer/Indicative content	Marks	Part marks and guidance	
18		a	DR $z = 2 \cos \frac{\pi}{6} + 2i \sin \frac{\pi}{6}$ $= \sqrt{3} + i$	M1 (AO1.1a) A1 (AO1.1) [2]		
		b	DR $z^2 = (\sqrt{3} + i)^2 = 3 + 2\sqrt{3}i - 1$ $= 2 + 2\sqrt{3}i$ OR $z = \left(2, \frac{\pi}{6}\right) \Rightarrow z^2 = \left(4, \frac{\pi}{3}\right) = 4\left(\cos \frac{\pi}{3} + i \sin \frac{\pi}{3}\right)$ $= 2 + 2\sqrt{3}i$	M1 (AO1.1a) A1ft (AO1.1) M1 A1ft [2]	for(their z) ² FT their z FT their z	
		c	DR $z^* = \sqrt{3} - i$ $\frac{z}{z^*} = \frac{\sqrt{3} + i}{\sqrt{3} - i} = \frac{\sqrt{3} + i}{\sqrt{3} - i} \times \frac{\sqrt{3} + i}{\sqrt{3} + i} = \frac{1}{2} + \frac{1}{2}\sqrt{3}i$ OR $z = \left(2, \frac{\pi}{6}\right) \Rightarrow z^* = \left(2, -\frac{\pi}{6}\right)$ $\Rightarrow \frac{z}{z^*} = \left(\frac{2}{2}, \frac{\pi}{6} - -\frac{\pi}{6}\right) = \left(1, \frac{\pi}{3}\right) = \left(\cos \frac{\pi}{3} + i \sin \frac{\pi}{3}\right)$ $\frac{1}{2} + \frac{\sqrt{3}}{2}i$	B1ft (AO1.1a) M1 (AO1.1) A1 (AO1.1) B1ft M1 A1 [3]	FT their z FT their z	Or $zz^* = z ^2$ (B1) and $\frac{z^2}{ z ^2} = \frac{2 + 2\sqrt{3}i}{4} = \frac{1}{2} + \frac{1}{2}\sqrt{3}i$
			Total	7		



Question			Answer/Indicative content	Marks	Part marks and guidance		
19		a	$(6 + 5i)(7 + 5i) = 6 \times 7 + 6 \times 5i + 5i \times 7 + 5 \times 5 \times -1$ $17 + 65i$	M1 (AO 1.1a) A1 (AO 1.1) [2]	Expanding brackets with $i^2 = -1$	If no working M0A0	
		b	$(6 - 5i)(7 - 5i) = 17 - 65i$ $(17 + 65i)(17 - 65i) = 17^2 + 65^2 = 4514$ $(6 + 5i)(6 - 5i) \text{ \& } (7 + 5i)(7 - 5i)$ $4514 = 61 \times 74$ so expressed as a product of prime factors $4514 = 2 \times 37 \times 61$	B1ft (AO 1.1) M1 (AO 3.1a) M1 (AO 3.1a) A1 (AO 3.2a) [4]	Conjugate of (a) Products considered separately Answer with no working M0M0A0.	Method using complex numbers must be shown.	
Total				6			

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Question	Answer/Indicative content	Marks	Part marks and guidance
20	DR Let $\sqrt{24+10i} = a + bi$ where a and b are real numbers. Then $24 + 10i = (a + bi)^2 = a^2 - b^2 + 2abi$ $\Rightarrow a^2 - b^2 = 24$ and $2ab = 10$ $\Rightarrow b = \frac{5}{a} \Rightarrow a^2 - \frac{25}{a^2} = 24$ $\Rightarrow a^4 - 24a^2 - 25 = 0$ $\Rightarrow (a^2 + 1)(a^2 - 25) = 0 \Rightarrow a = \pm 5, b = \pm 1$ $\Rightarrow (5 + i)$ and $-(5 + i)$ OR Square root of (r, θ) is $\left(\sqrt{r}, \frac{\theta}{2}\right)$ where $r = 26,$ $\theta = \tan^{-1} \frac{10}{24} = 22.62^\circ$ $\Rightarrow \pm(\sqrt{26}, 11.31) = \pm(\sqrt{26} \cos 11.31 + \sqrt{26} \sin 11.31i)$	M1 (AO3.1a) A1 (AO1.1) M1 (AO2.2a) A1 (AO1.1) A1 (AO2.2a) M1 A1 M1 A1 A1 [5]	



Question			Answer/Indicative content	Marks	Part marks and guidance	
			$= \pm (5 + i)$ aaaaaa			
			Total	5		



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Question			Answer/Indicative content	Marks	Part marks and guidance	
21		a	DR $e^{i\theta} - e^{-i\theta} = (\cos \theta + i \sin \theta)(\cos \theta - i \sin \theta) = 2i \sin \theta.$	B1 (AG) (AO 2.1) [1]	Must be clear evidence of use of Euler's equation and both $\cos(-\theta) = \cos \theta$ and $\sin(-\theta) = -\sin \theta$ seen or clearly implied.	eg just $(\cos(-\theta) + i \sin(-\theta))$ is not sufficient.
		b	DR $\frac{2e^{-i\theta}}{e^{i\theta} - e^{-i\theta}} \text{ or } \frac{2}{e^{i\theta}(e^{i\theta} - e^{-i\theta})}$ $= \frac{2ie^{-i\theta}}{2\sin \theta}$ $= \frac{i(\cos \theta - i \sin \theta)}{\sin \theta} = -(1 + i \cot \theta)$	B1 (AO 2.1) M1 (AO 2.1) A1 (AO 2.1) [3]	Condone omission of 2 Use of (a) and $1/i = -i$. Condone incorrect 2. AG At least one intermediate step must be shown.	



Question			Answer/Indicative content	Marks	Part marks and guidance	
		c	DR $C + iS = \sum 2e^{i\frac{\pi}{10}k}$ GP with and $r = e^{i\frac{\pi}{10}}$ seen or implied $\frac{2\left(\left(e^{i\frac{\pi}{10}}\right)^5 - 1\right)}{e^{i\frac{\pi}{10}} - 1}$ $\left(e^{i\frac{\pi}{10}}\right)^5 = i$ $\frac{2}{e^{i\frac{\pi}{10}} - 1} = -(1 + i \cot \frac{\pi}{20})$ $-(1 + i \cot \frac{\pi}{20})(i - 1) = \cot \frac{\pi}{20} + 1 + i(\cot \frac{\pi}{20} - 1)$ $C = \cot \frac{\pi}{20} + 1$	B1 (AO 3.1a) B1 (AO 2.2a) M1 (AO 2.2a) A1 (AO 1.1) B1 (AO 1.1) M1 (AO 2.2a) M1 (AO 3.1a) A1 (AO 2.2a) [8]	Expressing $C + iS$ as the sum of (multiples of) positive integer powers of $e^{i\frac{\pi}{10}}$ Could be implied by eg formula even if a and/or n incorrect. Use of formula for their series. All correct Correct use of (b) for their expression. Expanding brackets and collecting	



Question			Answer/Indicative content	Marks	Part marks and guidance		
		d	DR $S = \cot \frac{\pi}{20} - 1 = \cot \frac{\pi}{20} + 1 - 2 = C - 2$ All of the trigonometric terms are, in effect, the same; eg $\cos \frac{\pi}{10} = \sin \frac{2\pi}{5}$.	B1 (AO 2.2a) B1 (AO 2.4) [2]	Must be from fully correct working and solution in (c). Must be at least one explicit example or an appeal to the statement $\cos\left(\frac{\pi}{2} - \theta\right) = \sin \theta$ e.		
			Total	14			
22		a	$2^2 + 2 \times 2 \times i + i^2$ $= 4 + 4i - 1 = 3 + 4i$	M1 (AO 1.1a) A1 (AO 1.1) [2]	AG $i^2 = -1$ must be shown		
		b	Both z and z^2 correctly located	B1 (AO 1.1) [1]	If clearly labelled then award mark unless wildly inaccurate		
		c	$ z^2 = z ^2$.	B1 (AO 1.2) [1]			
		d	$\arg(z^2) = 2\arg(z)$.	B1 (AO 1.2) [1]			
			Total	5			



Question			Answer/Indicative content	Marks	Part marks and guidance		
23		i	$2 \cos \theta = e^{i\theta} + e^{-i\theta}$ $2^6 \cos^6 \theta = e^{6i\theta} + 6e^{4i\theta} + 15e^{2i\theta} + 20 + 15e^{-2i\theta} + 6e^{-4i\theta} + e^{-6i\theta}$ $2^6 \cos^6 \theta = (e^{6i\theta} + e^{-6i\theta}) + (6e^{4i\theta} + 6e^{-4i\theta}) + (15e^{2i\theta} + 15e^{-2i\theta}) + 20$ $\Rightarrow 64 \cos^6 \theta = 2 \cos 6\theta + 6(2 \cos 4\theta) + 15(2 \cos 2\theta) + 20$ $\Rightarrow \text{result}$	M1 A1 M1 A1 [4]	Expand $(e^{i\theta} + e^{-i\theta})^6$ for converting to multiple angles Complete argument including pairing up of e.g. terms in z^4 and z^{-4}	Must equate	

Examiner's Comments
A good number of candidates scored full marks on this part. Marks were lost by those who omitted to write equalities between successive lines, or did not show the grouping of exponentials into pairs, or who worked exclusively with z without ever defining it as $e^{i\theta}$. The best answers started with the standard formula

$$\cos \theta = \frac{e^{i\theta} + e^{-i\theta}}{2}$$

and then defined z to aid conciseness. Some candidates still insist on expanding binomials longhand when it is expected that they should use the formula.



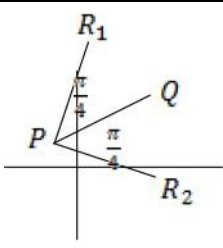
Question			Answer/Indicative content	Marks	Part marks and guidance		
		ii	$\cos 6\theta + 6 \cos 4\theta + 2 \cos 2\theta = 3$ $\Rightarrow \cos 6\theta + 6 \cos 4\theta + 15 \cos 2\theta + 10 = 3 + 13 \cos 2\theta + 10$ $\Rightarrow 32 \cos^6 \theta = 13(1 + \cos 2\theta)$ $\Rightarrow 32 \cos^6 \theta = 13(2 \cos^2 \theta)$ $\Rightarrow \cos \theta = 0$ or $\cos^4 \theta = \frac{13}{16}$ $\theta = \frac{1}{2}\pi, 0.319, 2.82$	M1* A1 *M1dep A1 A1 [5]	Use result from (i) Oe simplified form Use double angle identity		Examiner's Comments The majority of candidates successfully employed the identity from 7(i), together with the double-angle formula, to convert the equation to $32 \cos^6 \theta = 26 \cos^2 \theta$. From that stage onwards there was a divide between strong answers that tackled general equation-solving well and those that made standard errors. One error was to lose a solution by dividing through by $\cos^2 \theta$ rather than taking it out as a factor. The other common error was failing to find the two solutions in the range to $\cos^4 \theta = \frac{13}{16}$ due to neglecting the negative fourth root.
			Total	9			



Question			Answer/Indicative content	Marks	Part marks and guidance	
24		i	$ 2e^{\pi i/3}z = 2 z = 10$ $\text{Area} = \frac{1}{2} \cdot 10 \cdot 5 \cdot \sin \frac{1}{3}\pi$ $= \frac{25}{2}\sqrt{3}$	B1 M1 A1 [3]	Or $ 2e^{\pi i/3} = 2$ and scale area at end Use of formula with correct angle Soi by argand diagram Or $1/2bh$ since right angled triangle (21.7 inexact)	

Examiner's Comments
This question was generally well answered. Most candidates attempted to use the $\frac{1}{2}ab \sin C$ formula with the correct angle. Some successfully used $\frac{1}{2} \text{base} \times \text{height}$ (although they tended to sketch an argand diagram with real z which simplified their task). Common errors were to have sides of 5 and 5, rather than 5 and 10, or to use the complex numbers, rather than their moduli, for the sides.



Question			Answer/Indicative content	Marks	Part marks and guidance		
		ii	 $w = -1 + i + (4 + 2i)e^{\pm i\pi/4}$ $= \sqrt{2} - 1 + (3\sqrt{2} + 1)i$ or $3\sqrt{2} - 1 + (1 - \sqrt{2})i$	M1 A1 M1 A1 A1 [5]	Argand diagram with P, Q and attempt at one R at approx imately $\frac{\pi}{4}$ to PQ Diagram all correct SC1 if zero scored out of final 3 marks, for $(4 + 2i)e^{\pm i\pi/4} = \sqrt{2} + 3\sqrt{2}i$ or $3\sqrt{2} - \sqrt{2}i$	Including points labelled, angles labelled or R's in correct quadrant. Distances of Q and R's from P appear equal and gradients approximately correct condone omission of $\pm$ at M1 stage 0.41 + 5.24i 3.24 - 0.41i	

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Examiner's Comments

Some good Argand sketches were seen, with the best ones having realistic gradients for each PR and showing the arc, entered P, through Q. However, very many were untidy, with R_1 and R_2 poorly positioned. The lengths of PR_1 and PR_2 were often very different from that of PQ and the angles drawn were frequently far less or far



Question			Answer/Indicative content	Marks	Part marks and guidance	
					more than $\pi/4$. The second part as badly done by many candidates, with some failed attempts at using trigonometry. The best answers converted rotation and translation into complex algebra. Some candidates scored partial credit for finding $(4 + 2i)e^{\pm i\pi/4}$, but failed to complete the argument by adding $-1 + i$. One alternative approach which candidates often couldn't complete involved writing $\overline{PQ}$ in the form $Re^{i\theta}$ before adding $\pm \frac{\pi}{4}$ to the argument. Some candidates simply, erroneously, multiplied the modulus of $\overline{PQ}$ by $e^{\pm \frac{\pi}{4}}$.	
			Total	8		

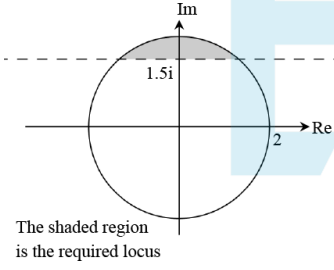


Question			Answer/Indicative content	Marks	Part marks and guidance		
25		a	(i) All roots satisfy $ z^5 = a^5 \Rightarrow z = a$	E1(AO 2.2a) [1]	Clear justification that all points are equidistant from O		
		a	(ii) All points are distance a from O Each root of the form $a \operatorname{cis}\left(\frac{2k\pi}{5}\right)$ hence spaced at angular intervals of $\frac{2\pi}{5}$ around this Circle	B1(AO 1.1) E1(AO 2.4) [2]	Using the answer to part (ii)		
		a	(iii) $\arg(b \times e^{\frac{2\pi i}{5}}) = \arg(b) + \arg(e^{\frac{2\pi i}{5}})$ $= \frac{4}{5}\pi = \arg(c)$	E1(AO 2.2a) [1]	Multiplying any complex number by $e^{\frac{2i\pi}{5}}$ rotates it through an angle of $\frac{2\pi}{5}$, so $b \rightarrow$		



Question			Answer/Indicative content	Marks	Part marks and guidance	
		a	(iv) $b^* = \operatorname{cis}\left(\frac{2\pi}{5}\right)^* = \operatorname{cis}\left(\frac{-2\pi}{5}\right) = \operatorname{cis}\left(\frac{8\pi}{5}\right) = e$	E1(AO 2.4) [1]	Or by noting that complex roots of a real polynomial occur in conjugate pairs (the pairing is obvious)	
		a	(v) a, b, c, d and e are the roots of $z^5 - a^5 = 0$ so sum of a, b, c, d and e is $-\frac{0}{1} = 0$	M1(AO 2.1) E1(AO 1.1) [2]	Accept any complete justification	



Question			Answer/Indicative content	Marks	Part marks and guidance		
		b	All roots of equal magnitude $a \cos \frac{\pi}{5}$ All roots spaced at angles $\frac{2\pi}{5}$ of $\frac{2\pi}{5}$ r is negative real root $\Rightarrow z^5 + (a \cos \frac{\pi}{5})^5 = 0 \Rightarrow 0^5 = 0$	M1(AO 3.1a) B1(AO 3.1a) A1(AO 2.2a) [3]	From any sensible right-angled triangle Must justify equally spread arguments Allow correct eqn. from 1st M1 only		
			Total	10			
26			 The shaded region is the required locus	B1(AO 1.1) B1(AO 1.1) B1(AO1.1) [3]	Circle centre 0, radius 2 Line parallel to x axis... Through 1.5i Correct shading with continuous arc and dashed line	Must state which region is the required locus	
			Total	3			



Question			Answer/Indicative content	Marks	Part marks and guidance		
27		a	DR $\alpha = 2 + 3i \Rightarrow \beta = 2 - 3i$ $\Rightarrow x^2 - 4x + 13 = 0$ is a quadratic factor $\Rightarrow f(x)$ $= (x^2 - 4x + 13)(x^2 + 6x + 13)$	B1(AO 2.2a) M1(AO 1.1) M1(AO 1.1) A1(AO 1.1) [4]	Attempt to create first quadratic Attempt to derive second quadratic	Must come from $(x - \alpha)(x - \beta)$	
		b	DR $\Rightarrow$ Roots are $2 \pm 3i, -3 \pm 2i$	A1(AO 1.1) [1]	Depends on M2 awarded in part (i)		
			Total	5			
28			$\frac{5}{2-4i} = \frac{5}{2-4i} \times \frac{2+4i}{2+4i} = \frac{10+20i}{20} = \frac{1}{2} + i$	M1(AO 2.1) A1(AO 1.1) [2]	For multiplying numerator and denominator by $2 + 4i$ Clear demonstration	or multiplying	
			Total	2			



Question		Answer/Indicative content	Marks	Part marks and guidance		
29		Let $C = \sum_{r=0}^{20} \binom{20}{r} \cos r\theta$ and $S = \sum_{r=1}^{20} \binom{20}{r} \sin r\theta$ $C + iS =$ $1 + 20(\cos \theta + i \sin \theta) + \binom{20}{2}(\cos 2\theta + i \sin 2\theta) +$ $\dots + (\cos 20\theta + i \sin 20\theta)$ $= 1 + 20z + \binom{20}{2}z^2 + \dots + z^{20}$ where $z = (\cos \theta + i \sin \theta)$ $= (1 + z)^{20}$ $\left(1 + 2\cos^2 \frac{\theta}{2} - 1 + i2\sin \frac{\theta}{2} \cos \frac{\theta}{2}\right)^{20}$ $\left(2\cos \frac{\theta}{2}\right)^{20} \left(\cos \frac{\theta}{2} + i \sin \frac{\theta}{2}\right)^{20}$ $(\cos 10\theta + i \sin 10\theta)$ Re: $C = 2^{20} \cos^{20} \left(\frac{1}{2}\theta\right) \cos 10\theta$	M1(AO2.1) M1(AO3.1a) A1(AO2.2a) A1(AO1.1) B1(AO3.1a) M1(AO3.1a) M1(AO2.2a) E1(AO3.2a) [8]	by de Moivre's theorem by de Moivre's theorem AG, must see the mention of taking the real part only.		
Total			8			

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Question			Answer/Indicative content	Marks	Part marks and guidance	
30		i	$ z = \sqrt{5}$	B1	Allow 2.2	
		i	$\arg z = -26.6^\circ$ or -0.464	B1	Allow -27° or $-0.46(3)$ Examiner's Comments Most found $ z $ correctly, but many candidates omitted a necessary negative sign in their value for $\arg z$.	
		ii	$a + b = 2, b - a = -8$ $a = 5, b = -3$	B1	$z^* = 2 + i$ stated or used	
		ii		M1	Obtain two equations from real and imaginary parts	
		ii		A1	Obtain correct equations	
		ii		M1	Attempt to solve 2 linear equations	
		ii		A1	Obtain correct answers Examiner's Comments The conjugate was generally given as $2 + i$, but a few candidates used $x + iy$ and $x - iy$ for z and z^* . Arithmetic slips in obtaining two linear equations or in solving their linear equations caused the most frequent loss of marks.	
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Question			Answer/Indicative content	Marks	Part marks and guidance	
31		i		B1	Circle	
		i		B1	Centre O and radius 2	
		ii		B1	Horizontal line	
		ii		B1	(3, 1) on their line	
		ii		B1	$\frac{1}{2}$ line to left i.e. horizontal	
					Examiner's Comments	
					Many candidates did not give sufficient detail in their sketches. The circle often had no clear indication of the radius, and the point (3,1) was not shown on the line or half-line. Also, what may have been meant to be a half-line was often drawn with a portion to the right of (3, 1). Some candidates thought that both loci were circles and some that both were straight lines.	
		iii		B1	Shade only inside their circle or above their horizontal line	
		iii		B1	Completely correct diagram	
					Examiner's Comments	
					Most candidates realised that the inside of the circle was required, but the second inequality was often not interpreted as meaning that the portion above the line was to be shaded.	
			Total	7		



Question			Answer/Indicative content	Marks	Part marks and guidance	
32		i	$z = 1, e^{2\pi i/5}, e^{4\pi i/5}, e^{6\pi i/5}, e^{8\pi i/5}$	M1	$e^{2\pi i/5}$ soi	Examiner's Comments Most candidates were clearly well versed in finding roots of unity. It is expected that the real root, 1, should not be left in exponential form. Those losing marks had sometimes misunderstood the instruction to give all non-real numbers in the given form resulting in them losing a mark due to only giving 4 roots. A few others completely omitted i in the exponents consequently scoring zero.
		i		A1		
		ii	$z^5 = -32$ has a root -2 , so roots are $-2, -2e^{2\pi i/5}, -2e^{4\pi i/5}, -2e^{6\pi i/5}, -2e^{8\pi i/5}$	M1	Use part (i) or from scratch	
		ii	Roots $-2, 2e^{7\pi i/5}, 2e^{9\pi i/5}, 2e^{\pi i/5}, 2e^{3\pi i/5}$	A1	cao with $r > 0, 0 < \theta < 2\pi$ (allow $2e^{\pi i}$ for -2)	
		ii	Argand diagram	M1	one root in each quadrant plus one on real axis	

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Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii		A1	axes and roots labelled. Roots equal moduli and equiangular spacing	Examiner's Comments The best answers gave the roots in precisely the required form and then produced 5 equally spaced, labelled points or vectors on an Argand diagram with labelled axes (Re, Im). The majority of candidates gained at least M1 for -2 times their roots of unity, and many successfully converted to the polar form with $r = 2$. Some, erroneously, simply tried 2 times the roots of unity or, realising that they had to deal somehow with the -32, used $2i$ times the roots of unity. Most candidates gained 1 of the 2 marks available for their Argand diagrams due to at least having located their roots in each quadrant; however, many were scrappy, with badly unequal lengths, angles or with asymmetric complex conjugate roots.
			Total	6		

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Question			Answer/Indicative content	Marks	Part marks and guidance	
33		i	$\cos 6\theta + i \sin 6\theta = (\cos \theta + i \sin \theta)^6$	B1	Use de Moivre	or $\sin 6\theta = \text{Im}(\cos \theta + i \sin \theta)^6$
		i	$= \cos^6 \theta + 6i \sin \theta \cos^5 \theta - 15 \sin^2 \theta \cos^4 \theta - 20i \sin^3 \theta \cos^3 \theta + 15 \sin^4 \theta \cos^2 \theta + 6i \sin^5 \theta \cos \theta - \sin^6 \theta$	B1	All terms correct	
		i	$\sin 6\theta = 6 \sin \theta \cos^5 \theta - 20 \sin^3 \theta \cos^3 \theta + 6 \sin^5 \theta \cos \theta$	M1	Compare imaginary parts	
		i	$= \cos \theta (6 \sin \theta (1 - \sin^2 \theta)^2 - 20 \sin^3 \theta (1 - \sin^2 \theta) + 6 \sin^5 \theta)$	M1	Take out factor of $\cos \theta$ and give other factor in terms of $\sin \theta$ only	



Question			Answer/Indicative content	Marks	Part marks and guidance	
		i	$= \cos \theta (32 \sin^5 \theta - 32 \sin^3 \theta + 6 \sin \theta)$	A1	ag Convincingly shown, having been explicit about taking imaginary parts	must have $\sin 6\theta = \dots$ 'final line' Examiner's Comments Where candidates produced good solutions, these were well laid out, with precise use of the equals sign; their scripts clearly referenced the application of De Moivre, showed their full binomial expansion, were explicit about taking the imaginary part, and showed how they replaced terms in cosine with equivalent ones in sine. While the use of c and s is acceptable in intermediate calculations, the letters should be defined at the start and not retained in the final answer. Many candidates failed to include this level of detail, whilst still accessing some of the marks available. Centres should note that conventionally (as given in Appendix B - Mathematical Notation) $\text{Im}(z) = y$ and not iy. Where candidates misapplied this convention, they were given the benefit of the doubt on this occasion.
		ii	$\frac{\sin 6\theta}{\sin 2\theta} = \frac{\cos \theta (32 \sin^5 \theta - 32 \sin^3 \theta + 6 \sin \theta)}{2 \sin \theta \cos \theta}$	M1	Complete the square	M1A1 for showing stationary points occur when $\sin^2 \theta = 0, 1$ or $\frac{1}{2}$
		ii	$= 16 \sin^4 \theta - 16 \sin^2 \theta + 3$	M1		
		ii	$= 4 (2 \sin^2 \theta - 1)^2 - 1$	A1		



Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	$\therefore \frac{\sin 6\theta}{\sin 2\theta} \geq -1$	M1	deduces lower bound	if using calculus, must convince for nature of stationary points for each M1 here
		ii	$0 \leq 2 \sin^2 \theta \leq 2$ $\therefore (2 \sin^2 \theta - 1)^2 \leq 1$	M1	deduces upper bound	can omit line 1 or 2 from the workings here, but not for final A mark
		ii	$\therefore 4(2 \sin^2 \theta - 1)^2 - 1 \leq 3$ $\Rightarrow -1 \leq \frac{\sin 6\theta}{\sin 2\theta} \leq 3$		SC if none of marks 2 to 5 (M1A1M1M1) gained then SC M1A1 for any valid method of deducing upper bound, and similarly for lower bound	

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Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	But upper bound attained $\Rightarrow$ $\sin^2 \theta = 0$ or 1 $\Rightarrow \sin 2\theta = 0$	M1	Dep on showing valid method for $UB \leq 3$	Or independent proof that not equal to 3 Examiner's Comments Most good solutions to this question resulted from completing the square or from simplifying the function into one involving multiple angles. Once either of these steps was taken, candidates were much more capable of convincing with their justifications. Many made some progress by differentiating with respect to θ and equating to zero, but then usually failed to find all stationary points or to convince on their nature. Still others made the substitution $x = \sin \theta$ and then treated the expression as a quadratic, but then failed to take account of the extra complexities due to the range of x being restricted to $[1, -1]$. Another approach which candidates used when trying to find the minimum was to consider the intersection of the function with the line $y = k$, and finding where the discriminant of the resulting equation was non-negative. Many candidates merely worked back from the given answer, and scored only the first mark (for using $\sin 2\theta = 2 \sin \theta \cos \theta$ to simplify the initial expression). Some candidates equated the function to zero for some reason and then factorised this expression, to no real benefit.



Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	$\Rightarrow -1 \leq \frac{\sin 6\theta}{\sin 2\theta} \leq 3$ So $\sin 2\theta \neq 0$	A1	full convincing overall argument	
			Total	12		



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Question			Answer/Indicative content	Marks	Part marks and guidance	
34		i	Diagram	B1		must have triangle where B is anticlockwise from A, looks isosceles, $\angle AOB < \frac{\pi}{4}$, if axes labelled then must be correct
		i	$OB = ze^{\frac{1}{6}\pi i} = z e^{\frac{1}{6}\pi i} = z \cdot 1 = z = OA$	M1		condone $OB = z = OA$
		i	So triangle is isosceles or	A1	without contradictions	



Question			Answer/Indicative content	Marks	Part marks and guidance	
		i	$\angle AOB = \frac{1}{6}\pi$	B1	or 30° <u>Examiner's Comments</u> Good Argand diagrams were drawn with attention to detail so that axes were labelled but not scaled, angle AOB was clearly less than 45° OA and OB looked to be the same length and the triangle was clearly neither equilateral nor right angled. The majority of candidates started by placing A on the x axis which is, of course, a quite permissible location; but by doing so, they tended to lose the sense of it being a general complex number. As a result, in attempting to show that the triangle was isosceles, it was common to see candidates claim that the two moduli or lengths were equal to z (rather than $ z $). Some only considered the specific case $ z = 1$. A number of others believed that what was required was to demonstrate that the triangle was not equilateral in order to fully justify their conclusion. Centres should stress the hierarchical nature of the subject means that isosceles includes equilateral. There was also a number of candidates who attempted to calculate the area of their triangle – quite why this occurred is unclear.	Can be just on diagram
		ii	$w = (1+i) + ((5+2i) - (1+i))e^{\pm \frac{1}{6}\pi i}$	M1	Rotation of CD	Condone omission of $\pm$ in M marks
		ii		M1	Translation of attempted CE	



Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii		A1		
		ii	$w = \frac{1}{2} + 2\sqrt{3} + \left(3 + \frac{1}{2}\sqrt{3}\right)i$	M1	converts $e^{\pm \frac{1}{6}\pi i}$ into $a + bi$ form	
		ii	or $\frac{3}{2} + 2\sqrt{3} + \left(-1 + \frac{1}{2}\sqrt{3}\right)i$	A1		
		ii	Alternative method: $CE = \begin{pmatrix} a \\ b \end{pmatrix}, CD = \begin{pmatrix} 4 \\ 1 \end{pmatrix}$. Now use $CE \cdot CD = 17\cos(\pi / 6)$ and $CE^2 = 17$ to obtain equations $4a + b = 17\sqrt{3} / 2$ and	M1	(for both).	
		ii	$a^2 + b^2 = 17$ (or equivalent)	A1		
		ii	Obtain 3-term quadratic in one variable and solve to get one correct value of a or b	M1	Quadratics are $a^2 - 4\sqrt{3}a + 47 / 4 = 0$ and $b^2 - \sqrt{3}b - 13 / 4 = 0$	
		ii	$(a, b) = (2\sqrt{3} \pm \frac{1}{2}, \frac{1}{2}\sqrt{3} \mp 2)$	A1		



Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	Final answer	A1	<u>Examiner's Comments</u> Strong candidates produced neat and concise solutions. Often the best solutions had a clear grasp of the relationship between vector methods and the Argand diagram. Others obtained good marks by meandering methods that sought to evaluate anything and everything that might be useful until an appropriate solution was found, even if this was only as a decimal approximation rather than, as required, the exact value. In weaker scripts, frequently the only mark picked up was that for the conversion of $e^{i\pi/6}$ to $a + bi$ form. Many tried to rotate $5 + 2i$ (rather than $4 + i$), or, having rotated $4 + i$, failed to translate the resulting expression. A number obtained an equation of the form $a^2 + b^2 = 17$ (where $w = a + bi$) but were unable to gain further information about a and b .	
			Total	9		

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Question			Answer/Indicative content	Marks	Part marks and guidance	
35		i	$\cos 4\theta + i \sin 4\theta = (\cos \theta + i \sin \theta)^4$		soi by at least	Can be broken down already but with <i>i</i> 's in place
		i		B1	$\tan 4\theta = \frac{im((\cos \theta + i \sin \theta)^4)}{re((\cos \theta + i \sin \theta)^4)}$	
		i	$= c^4 + 4ic^3 s - 6c^2s^2 - 4ics^3 + s^4$	B1		
		i	Taking re and im parts $\cos 4\theta = c^4 - 6c^2s^2 + s^4$			
		i	$\sin 4\theta = 4c^3s - 4cs^3$	M1	take real and imaginary parts AG. Must show division of numerator	
		i	$\tan 4\theta = \frac{4 \tan \theta - 4 \tan^3 \theta}{1 - 6 \tan^2 \theta + \tan^4 \theta}$	A1	and denominator by c^4 and must have been explicit about re and im	
					<u>Examiner's Comments</u> Where candidates produced good solutions, these were well laid out, with precise use of the equals sign. These scripts clearly referenced the application of De Moivre, showed their binomial expansion, were explicit about taking real and imaginary parts and made clear that they were dividing numerator and denominator by $\cos^4 \theta$ to get to the final identity required. Many candidates failed to include this level of detail, whilst still accessing some of the marks available.	
		ii	Rearranging polynomial gives			
		ii	$1 - 6t^2 + t^4 = \sqrt{3}(4t - 4t^3)$	M1		
		ii	so $\tan 4\theta = \frac{1}{\sqrt{3}}$	A1		



Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	$4\theta = \text{their } \frac{1}{6}\pi + n\pi$	B1		
		ii	$t = \tan \theta = \tan \frac{1}{24}\pi, \tan \frac{7}{24}\pi, \tan \frac{13}{24}\pi, \tan \frac{19}{24}\pi$			condone all angles seen and no extras, but t not given as equal to $\tan \theta$
		ii		B1	one correct	
		ii		B1	all correct or (4) equivalent	
					<u>Examiner's Comments</u> Many candidates were able to use the connection to part (i) to see that $\tan 4\theta = \frac{1}{\sqrt{3}}$ even if their method in getting there was often convoluted. Strong candidates were then able to find the roots with ease. Two common faults were either to presume that the roots were the four angles or to give more than four solutions. Solving $\tan 4\theta = \frac{1}{\sqrt{3}}$ is an AS Mathematics skill, yet some candidates only gave one solution, or were unfamiliar with the correct method for finding further solutions.	
			Total	9		



Question			Answer/Indicative content	Marks	Part marks and guidance	
36		i	$z^6 = 1 \Rightarrow z = e^{2k\pi i/6}$	M1		
		i	$k = 0,1,2,3,4,5$	A1	Oe exactly 6 roots	accept roots 1, -1 given as integers.
		i	Diagram	B1	6 roots in right quadrant,	
		i		B1	correct angles and moduli	as evidenced by labels, circles, or accurate diagram, or by co-ordinates
					Examiner's Comments	
					Although the majority of candidates could find the roots of unity, displaying them was not as well done. Candidates remain unwilling to give appropriate care and attention to Argand diagram sketches, resulting in lack of essential detail. Good answers were marked by clear labelling of axes and roots and an explicit reference within the Argand diagram to the polar coordinates of these roots. This could be (but didn't have to be) succinctly done by means of marked angles and a circle through all roots.	
	ii		$(1+i)^6 = \left(\sqrt{2}e^{\frac{1}{4}\pi i}\right)^6$	M1	Attempts modulus – argument form, getting at least 1 correct	
	ii		$8e^{\frac{3}{2}\pi i}$	M1	for $(\text{mod})^6$ and $\arg \times 6$	
	ii		$= -8i$	A1		complete argument including start line
	ii		Alternative: $(1+i)^6 = 1 + 6i + 15i^2 + 20i^3 + 15i^4 + 6i^5 + i^6$	M1		
	ii		$= 1 + 6i - 15 - 20i + 15 + 6i - 1$	M1	no more than 1 term wrong	Sc 2 for only lines 2 & 3 correct
	ii		$= -8i$	A1		
	ii		Alternative: $(1+i)^2 = 2i$	M1		



Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	$(1 + i)^6 = (2i)^3$	M1	ag	
		ii	$= -8i$	A1		
					Examiner's Comments	
					Candidates' solutions were split fairly evenly between the 3 approaches given in the markscheme. Arguments were sometimes insufficiently detailed; so those using the binomial expansion method occasionally neglected to show an unsimplified stage. Candidates at this level are expected to structure their demonstrations and proof, so those who did not include " $(1+i)^6 =$ " were not credited with the final accuracy mark.	
		iii	$z^6 = -8i \Rightarrow z = (1 + i)e^{2k\pi i/6}$	M1	or equivalent k	
		iii	$= \sqrt{2}e^{i\frac{\pi}{4}}e^{2k\pi i/6}$	M1		
		iii	$\sqrt{2}e^{i\pi(1/4+k/3)}, k = 0,1,2,3,4,5$	A1		
		iii	Alternative: $z^6 = 8e^{i\pi(\frac{3}{2}+2k)}$	M1		
		iii	$\sqrt{2}e^{i\pi(1/4+k/3)}, k = 0,1,2,3,4,5$	M1		
		iii		A1		or equivalent: e.g. $\sqrt{2}e^{i\pi(-1/12+k/3)}$ accept unsimplified modulus



Question			Answer/Indicative content	Marks	Part marks and guidance	
		iii			Examiner's Comments This question was well answered by strong candidates, but many were unable to make any headway with it. Frequently, candidates had more joy through approaching the question independently of the preceding parts, simply using the alternative method given in the markscheme.	
			Total	10		



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Question			Answer/Indicative content	Marks	Part marks and guidance	
37		i	$2i \sin \theta = e^{i\theta} - e^{-i\theta}$	B1	any equivalent form	If use z, must define it
		i	$2i \sin n\theta = e^{in\theta} - e^{-in\theta}$			
		i	$(2i \sin \theta)^5 = (e^{i\theta} - e^{-i\theta})^5$			
		i	$= e^{i5\theta} - 5e^{i3\theta} + 10e^{i\theta} - 10e^{-i\theta} + 5e^{-i3\theta} - e^{-i5\theta}$	M1*	binomial expansion	can be unsimplified
		i	$32i \sin^5 \theta = (e^{5i\theta} - e^{-5i\theta}) - 5(e^{3i\theta} - e^{-3i\theta}) + 10(e^{i\theta} - e^{-i\theta})$	M1*dep	grouping terms	Award B1 then sc M1A1 for candidates who omit this stage from otherwise complete argument
		i	$= 2i \sin 5\theta - 5(2i \sin 3\theta + 10(2i \sin \theta))$			
		i	$\sin^5 \theta = \frac{1}{16}(\sin 5\theta - 5 \sin 3\theta + 10 \sin \theta)$	A1	AG	must convince on the $\frac{1}{16}$ and on the elimination of i
					Examiner's Comments A significant number of candidates had not learned the basic result initially requested, and gave an expression for $\sin \theta$ which omitted "i". This then led to incorrect conclusions being drawn at the end of their working as they looked to "show" the given result. It was also concerning how many candidates both here and in Qu. 3(ii) seemed completely unaware of the formula for a binomial expansion, leading them to carry out lengthy, often error ridden, expansions of brackets. Centres would do well to stress that, in proofs of this nature, the binomial expansion formula can be directly applied. The best answers were precise and concise, but showed each step of the proof, which is a standard one.	
		ii	$16 \sin^5 \theta - 10 \sin \theta = \sin 5\theta - 5 \sin 3\theta$	M1*	Attempts to eliminate $\sin 5\theta$ and $\sin 3\theta$	



Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	$16\sin^5\theta - 6\sin\theta = 0$	A1		Or $16\sin^5\theta = 6\sin\theta$
		ii	$\sin\theta = 0, \pm\sqrt[4]{\frac{3}{8}}$	M1dep*	must have 3 values for $\sin\theta$	
		ii	$\theta = 0, \pm 0.899$	A1		
		ii			Examiner's Comments Few had problems in being able to rearrange the given equation to obtain $16\sin^5\theta = 6\sin\theta$, or equivalent but many then ignored the $\sin\theta = 0$ solution or, when fourth rooting, omitted the negative root. It was pleasing to see a few strong candidates considering the fact that they were solving a quintic and so were on the lookout for 5 solutions for $\sin\theta$ (two of which they observed to be imaginary). It is this ability to consider a problem analytically that forms the basis of the best approaches to the problem-solving aspect of the course.	
			Total	8		

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Question			Answer/Indicative content	Marks	Part marks and guidance	
38		i	Sketch			Must have axes, A shown 3 across and either scale (or co-ordinates)
		i		B1	with B in rough position, or angle and distance on argand diagram. No inconsistencies	
		i	$OA = 3 = 3, OB = 3e^{\frac{1}{3}\pi i} = 3$			
		i	and $\angle BOA = \frac{1}{3}\pi$	M1	Can be seen on diagram	Alt. Attempts AB or second angle
		i	hence $\triangle OAB$ equilateral	A1	Examiner's Comments	
					This question was a good differentiator, with each part producing a balanced spread of marks. Poor quality sketches were evident from many candidates, with many omissions of the key features necessary on an Argand diagram. This was often linked with limited evidence being given for establishing that the triangle was equilateral, or a failure to conclude arguments. Good answers often used clear notation to show how candidates were, for instance, finding the length of AB.	
		ii	$3e^{-\frac{1}{3}\pi i}$		Or $3e^{\frac{5}{3}\pi i}$. Isw	



Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii		M1A1	M1 for evidence they are considering BA, or for $\frac{3}{2} - \frac{3}{2}\sqrt{3}i$ Examiner's Comments This was generally done well with many completely correct solutions given. A number were, however, only able to express the result in cartesian rather than polar form. Exponential form, cis form and polar coordinate form were all acceptable, though candidates should be made aware that neither $3\cos(-\frac{1}{3}\pi) + 3i\sin(-\frac{1}{3}\pi)$ nor $3(\cos(\frac{1}{3}\pi) - i\sin(\frac{1}{3}\pi))$ is standard polar form, and these were penalised.	For full marks can use CiS form, or clear polar co-ordinates, in radians. Not C-is
		iii	$(3 - 3e^{\frac{1}{3}\pi i})^5 = 3^5 e^{-\frac{5}{3}\pi i}$	M1	For mod ⁵ and arg $\times 5$	"Hence" so must use 'their $3e^{-\frac{1}{3}\pi i}$ '
		iii	$= 243(\cos \frac{5}{3}\pi - i\sin \frac{5}{3}\pi)$	A1ft	aef	
		iii	$= \frac{243}{2} + \frac{243}{2}\sqrt{3}i$	B1	Examiner's Comments Candidates who had given part (ii) in polar form were usually able to use this to gain at least the first two marks. Since the question starts "Hence", those who tried to solve the question by means of a binomial expansion were only able to access the final mark; they were rarely successful in doing this.	Condone use of "121.5".
			Total	8		



Question			Answer/Indicative content	Marks	Part marks and guidance	
39		i	$\cos 5\theta + i \sin 5\theta = (\cos \theta + i \sin \theta)^5$		Or	
		i		B1		
		i			$\cos 5\theta = \operatorname{re}\{(\cos \theta + i \sin \theta)^5\}$	
		i	$= c^5 + 5ic^4s - 10c^3s^2 - 10ic^2s^3 + 5cs^4 + is^5$	M1		No more than 1 error, can be unsimplified
		i	$\cos 5\theta = c^5 - 10c^3s^2 + 5cs^4$	M1	Take real parts	
		i	$= c^5 - 10c^3(1 - c^2) + 5c(1 - c^2)^2$	M1		
		i	$= c^5 - 10c^3 + 10c^5 + 5c - 10c^3 + 5c^5$			



Question			Answer/Indicative content	Marks	Part marks and guidance	
		i	$\cos 5\theta = 16c^5 - 20c^3 + 5c$	A1	AG Examiner's Comments Although many excellent solutions were seen, a number failed to give sufficient working to truly show the given expression to be correct. These candidates commonly neglected to quote de Moivre's theorem at the start of their argument, or omitted the full binomial expansion. On occasions candidates crossed out terms as they transferred them from one line of their argument to the next. Although this may help these candidates keep track of terms, it does undermine their argument and could, potentially, be penalised. This practice should be discouraged. Terms should only be crossed out when they cancel out and even then it is important that these terms remain visible. A few tried to start from $\cos \theta = \frac{1}{2}(z + z^{-1})$, but hardly any made progress with this approach.	
		ii	Multiplying by x gives $16x^5 - 20x^3 + 5x = 0$			Hence, so no marks for using quadratic at this stage.
		ii	letting $x = \cos \alpha$ gives $\cos 5\alpha = 0$	M1		
		ii	hence $5\alpha = \frac{1}{2}\pi, \frac{3}{2}\pi, \frac{5}{2}\pi, \frac{7}{2}\pi, \frac{9}{2}\pi$ $\alpha = \frac{1}{10}\pi, \frac{3}{10}\pi, \frac{5}{10}\pi, \frac{7}{10}\pi, \frac{9}{10}\pi$	A1		



Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	$\cos \frac{5}{10}\pi = 0$ which is not a root	A1		
		ii	so roots $x = \cos \frac{1}{10}\pi, \cos \frac{3}{10}\pi, \cos \frac{7}{10}\pi, \cos \frac{9}{10}\pi$	A1	Examiner's Comments Many were confused about exactly what was required in this part and in part (iii). Candidates could not gain credit here for working out the surd value of the root and then using their calculators to find this in the form $\cos \alpha$. Of those who correctly considered solutions to $\cos 5\theta = 0$, many failed to rule out the value of $\theta = \frac{1}{2}\pi$, despite this not being a valid solution to the given quartic equation. A small number also failed to give solutions in the form $\cos \alpha$, appearing to believe that the angles themselves were the roots of the quartic.	
		iii	$16x^4 - 20x^2 + 5 = 0 \Leftrightarrow x^2 = \frac{20 \pm \sqrt{80}}{32}$	B1		Can be gained if seen in (ii)
		iii	cos decreases between 0 and π so $\cos \frac{1}{10}\pi$ is greatest root	M1		

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Question			Answer/Indicative content	Marks	Part marks and guidance	
		iii	$\text{so } \cos \frac{1}{10} \pi = \sqrt{\frac{20 + \sqrt{80}}{32}} = \sqrt{\frac{5 + \sqrt{5}}{8}}$	A1	Dep on full marks in (ii) Examiner's Comments The majority gained the B1 mark, sometimes from working in part (ii), but relatively few gave a satisfactory justification for picking the correct value for $\cos \frac{1}{10} \pi$. Since parts (ii) and (iii) both use the "hence" request, full marks could not be gained unless candidates' arguments over the two parts convinced the examiner that a credible line of reasoning had been followed. A few candidates tried to use sum and/or product of roots rules but this did not help.	
			Total	12		



Question			Answer/Indicative content	Marks	Part marks and guidance	
40		i	$1, e^{\frac{2\pi i}{5}}, e^{\frac{4\pi i}{5}}, e^{\frac{6\pi i}{5}}, e^{\frac{8\pi i}{5}}$	M1	Attempt roots	e.g. gives roots in an incorrect form.
		oe polar form				
		i		A1	<u>Examiner's Comments</u> Most candidates were familiar with this standard problem and many gained full marks. A variety of notation can be described as 'polar form', so it was acceptable to write the answer in exponential form, CiS form or in polar coordinates with argument in the interval $[0, 2\pi)$ or $(-\pi, \pi]$. Candidates were, however, penalised for using degrees, omitting i (commonly noted in exponential form attempts) or for roots of the form $\cos\theta - i\sin\theta$.	
	ii		$z^5 = (z + 1)^5 = z^5 + 5z^4 + 10z^3 + 10z^2 + 5z + 1$	M1		
	ii		$\Leftrightarrow 5z^4 + 10z^3 + 10z^2 + 5z + 1 = 0$	A1		
	ii		so $z + 1 = ze^{\frac{2k\pi i}{5}}$, $k = 0, 1, 2, 3, 4$	M1		
	ii		$k = 0$ no solution	B1	so i	



Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	$1 = z \left(e^{\frac{2k}{5}\pi i} - 1 \right)$ $z = \frac{1}{e^{\frac{2k}{5}\pi i} - 1}, k = 1, 2, 3, 4$	A1	If B0, then give A1 ft for correct solution plus $k = 0$ <u>Examiner's Comments</u> Although many candidates were able to tackle this question, few of them recognised that, with a quartic equation, there would only be four roots; consequently most candidates did not spot the fact that the value corresponding to $k = 0$ should be rejected.	
			Total	7		



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Question			Answer/Indicative content	Marks	Part marks and guidance	
41		i	(a)	M1	Sum of a GP	
		i	$e^{i\theta} + e^{2i\theta} + \dots + e^{10i\theta} = \frac{e^{i\theta} \left((e^{i\theta})^{10} - 1 \right)}{e^{i\theta} - 1}$	A1	AG	
		i	$= \frac{e^{\frac{1}{2}i\theta} (e^{10i\theta} - 1)}{e^{\frac{1}{2}i\theta} - e^{-\frac{1}{2}i\theta}}$	M1		
		i	$= \frac{e^{\frac{1}{2}i\theta} (e^{10i\theta} - 1)}{2i \sin(\frac{1}{2}\theta)}$	A1		
		i	(b) $\theta = 2n\pi \Rightarrow \text{sum} = 10$	B1	<u>Examiner's Comments</u> It was pleasing, in part (a), to see that most candidates spotted that they were dealing with a geometric progression and were able to gain the first two marks for this question. Converting $\frac{a(1-r^n)}{1-r}$ (given in the <i>List of Formulae</i> MF1) to the final form was beyond some candidates.	
		ii	$\cos \theta + \cos 2\theta + \dots + \cos 10\theta = \operatorname{Re} \left(\frac{e^{\frac{1}{2}i\theta} (e^{10i\theta} - 1)}{2i \sin(\frac{1}{2}\theta)} \right)$	M1	Take real parts	Must at least make genuine progress in sorting real part of numerator, or in converting numerator to trig terms.
		ii	$= \frac{\operatorname{Re} \left(-ie^{\frac{1}{2}i\theta} (e^{10i\theta} - 1) \right)}{2 \sin(\frac{1}{2}\theta)} = \frac{\operatorname{Re} \left(-ie^{\frac{21}{2}i\theta} + ie^{\frac{1}{2}i\theta} \right)}{2 \sin(\frac{1}{2}\theta)}$	M1	<u>Examiner's Comments</u> Part (b) was often attempted by using the formula in part (a) instead of realising the necessity to refer back to the original definition of S.	



Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	$= \frac{\sin\left(\frac{21}{2}\theta\right) - \sin\left(\frac{1}{2}\theta\right)}{2\sin\left(\frac{1}{2}\theta\right)}$ $= \frac{\sin\left(\frac{21}{2}\theta\right)}{2\sin\left(\frac{1}{2}\theta\right)} - \frac{1}{2}$	A1	AG	<u>Examiner's Comments</u> Those candidates who used the real part of (a) plus de Moivre's theorem could, with care concerning i in the denominator, produce the right-hand side from the left with relative ease. Some candidates, however, chose to use $\cos n\theta = \frac{z^n + z^{-n}}{2}$ for each term. If they were adept at algebraic manipulation, they could still work their way through to the solution by summing two geometric series. Some candidates, though, tended to make errors such as believing, mistakenly, that there was now one geometric series with $r = z - z^{-1}$.
		iii	$\cos\frac{1}{11}\pi + \cos\frac{2}{11}\pi + \dots + \cos\frac{10}{11}\pi = \frac{\sin\left(\frac{21}{22}\pi\right)}{2\sin\left(\frac{1}{22}\pi\right)} - \frac{1}{2}$	M1		For second M1, must convince that solution is exact and not simply from calculator.
		iii	But $\sin\frac{21}{22}\pi = \sin\left(\pi - \frac{21}{22}\pi\right) = \sin\frac{1}{22}\pi$	M1		



Question			Answer/Indicative content	Marks	Part marks and guidance	
		iii	So $\text{RHS} = \frac{1}{2} - \frac{1}{2} = 0$, so $\frac{1}{11}\pi$ is a root	A1	AG	
					<u>Examiner's Comments</u> Many candidates did not realise that it is insufficient to use (inexact) decimal calculator values for functions when trying to show the exact value of a root. Because they were not trying to solve the equation $\sin \frac{21\theta}{2} = \sin \frac{\theta}{2}$ they were then unable to make progress when it came to finding another root. Some candidates availed themselves of their knowledge of the identities $\sin x = \sin (\pi - x) = \sin (x + 2\pi)$ to produce neat, well argued solutions.	
		iii	Using $\sin(2\pi + x) = \sin x$ gives $2\pi + \frac{1}{2}\theta = \frac{21}{2}\theta \Rightarrow \theta = \frac{1}{5}\pi$	A1		
			Total	12		

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Question			Answer/Indicative content	Marks	Part marks and guidance	
42		i	$x = 2\sqrt{3} \cos(-\frac{\pi}{3}), y = 2\sqrt{3} \sin(-\frac{\pi}{3}), x^2 + y^2 = 12, y = -x\sqrt{3}$	M1	Correct trig expression for x or y allow a positive angle or 2 equations for x and y, not involving trig	
		i	$\sqrt{3} - 3i$	A1	Obtain correct answer as a complex number, extra answers not rejected gets A0 Examiner's Comments Those candidates who sketched an Argand diagram and used trigonometry usually found the correct value for z. Candidates who gave equations for the modulus and argument often made a sign error, not seeing that z is in the 4 th quadrant.	
		ii	$\sqrt{3} + 3i$	B1ft	Correct conjugate seen, ft from their z in (i)	
		ii		M1	Expand denominator	
		ii	$-1 - (4\sqrt{3})i$	A1	Correct value seen	
		ii		M1	Attempt to rationalise	
		ii	$-\frac{1}{49} + \frac{4\sqrt{3}}{49}i$	A1	Obtain correct answer a.e.e.f. N.B. if 2 answers given in (i) award marks for better solution in (ii) Examiner's Comments The conjugate and the method of rationalising was shown by most candidates, but again errors in arithmetic were frequent, with e.g. the value of the denominator, 49, being found as 47.	
			Total	7		



Question			Answer/Indicative content	Marks	Part marks and guidance	
43		i	$3 + 3i$	B1	Obtain centre as a complex number, allow (3, 3) but not (3, 3i), give if correct in locus equation	
		i	$\sqrt{5}$	B1	Obtain correct radius a.e.f. condone decimals	
		i	$ z - 3 - 3i = \sqrt{5}$	M1	Use correct form for locus	
		i		A1ft	Obtain correct answer from their centre and radius, $z = x + iy$ is acceptable Examiner's Comments There was a certain amount of confusion between the Cartesian and complex forms here. The centre was often written as (3, 3i), while the complex form of the circle was given as $(x - 3)^2 + (y - 3i)^2 = r^2$.	
		ii	Circle, centre in 1st quadrant, not touching or intersecting axes	B1	Centre need not be (3, 3)	
		ii	Straight line with -ve slope through (3, 3)	B1	(3, 3) may be implied by working from (i) must be longer than the diameter. N.B. No circle drawn, B1 for / may be earned Examiner's Comments The sketching was not particularly good, omission of any scale or coordinates was quite frequent, while the circle often crossed either or both of the axes. The line / was often drawn at the ends of the diameter, rather than through the centre of the circle.	



Question			Answer/Indicative content	Marks	Part marks and guidance	
		iii	$2 + 5i$ $4 + i$	M1	Gradient of l is -2 used or attempt to solve Cartesian equations for C and l M0 if equations for C and/or l contain i .	
		iii		A1 A1	Obtain correct answers, must be complex numbers Examiner's Comments Many candidates found the Cartesian equations of C and l correctly and then solved to find the correct points of intersection as coordinates, but then failed to give these points as complex numbers. Those who had drawn a good sketch were often able to find the points of intersection correctly with little effort.	
			Total	9		



Question			Answer/Indicative content	Marks	Part marks and guidance	
44		i	$x^2 - y^2 = 9, xy = 20$	M1	Equate real and imaginary parts of $9 + 40i$ and $(x + iy)^2$	
		i		A1	Obtain correct equations a.e.f., e.g. allow $2ixy = 40i$	
		i		M1	Rearrange to obtain quadratic in x^2 or y^2	
		i		M1	Solve quadratic in x^2 or y^2 and square root	
		i		A1	Obtain one correct answer as a complex number	
		i	$\pm(5 + 4i)$	A1	Obtain other answer as complex number a.e.f. $\pm 5 \pm 4i$ is OK for A2, $\pm(5 \pm 4i)$ scores A1 only	
					Examiner's Comments	
					Most candidates were able to show the correct method for finding the square roots of the given complex number. Sign errors caused the greatest loss of marks, and very few checked their answers by squaring the results that had been found, a simple process that would have highlighted an error in the working.	

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Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii		B1	Use sum and product of roots or solve equation or substitute into equation	
					Examiner's Comments	
					Insufficient working often caused the loss of this mark. Those who substituted into the given quadratic omitted to show the expanded version, while those who used the sum and product of roots approach often only used one of the two results. Errors in using the quadratic formula were also quite common.	
		iii	$u = \frac{1}{5 + 4i}$	B1	Use substitution to obtain one of the equations from other equation, (must have = 0)	
		iii	$\pm(\frac{5}{41} + \frac{4}{41}i), \pm(\frac{5}{41} - \frac{4}{41}i)$	M1	Use result from (i) (either root)	
		iii		A1	Obtain any 2 correct answers	

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Question			Answer/Indicative content	Marks	Part marks and guidance	
		iii		A1	Obtain other 2 correct answers S.C.: If they solve quadratic for u^2 directly, use scheme above, M1 when they have used (i) to get a value for u . Examiner's Comments This part proved quite taxing. Many did not show a convincing connection between the z and u equations. Only a small proportion of candidates realised that the quartic equation had four roots, the other pair being easily derived from the conjugate root, $9 - 40i$.	
			Total	11		
45			$z^* = x - iy$ $ z = \sqrt{x^2 + y^2}$ $2(x^2 + y^2)$	B1 B1 B1	Conjugate stated or used Modulus or it's square stated or used Obtain correct answer, a.e.f. but not involving i Examiner's Comments The conjugate was known by nearly all candidates, but there were many errors in the modulus. Some thought that $ z = \sqrt{2}$ or $x^2 + y^2$, while many expanded zz^* and obtained an expression with terms involving " i " or obtained $x^2 - y^2$.	
			Total	3		



Question			Answer/Indicative content	Marks	Part marks and guidance	
46		i		B1	Circle centre (-2, 0) or circle centre (2, 0)	
		i		B1	Touching y-axis at origin	
		i		B1	Half line with negative slope upwards	
		i		B1	Completely correct diagram	
					Examiner's Comments	
					Many did not see that the centre of the circle was on the x-axis and those who did often were not careful enough to make their circle touch the y-axis at the origin. The half line was usually correct, but a significant minority thought that $\frac{5\pi}{6}$ was an acute angle.	



Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	$-2 - \sqrt{3} + i$	B1ft	Correct real part and correct imaginary part of a complex number, ft for their half line from centre of their circle, allow decimals (-3.73 or better) or trig expressions Examiner's Comments Some tried to find the Cartesian equations of the loci and then solve them, usually with little success. Those who used a simple trigonometrical approach often made errors in the real part, $-\sqrt{3}$ instead of $(-2 - \sqrt{3})$. Many gave the answers as a pair of coordinates, thus not answering the question, which required a complex number.	
		ii		B1ft		
		iii		B1ft	Shade inside their circle	
		iii		B1	Completely correct diagram and shading S.C. allow last B1 for radius or complete line Examiner's Comments Most candidates recognised that the area inside the circle was required, but the correct sector was often not shown correctly.	
			Total	8		



Question			Answer/Indicative content	Marks	Part marks and guidance	
47		i		M1	Attempt to equate real and imaginary parts of $(x + iy)$ and $5 + 12i$	
		i	$x^2 - y^2 = 5, 2xyi = 12i$	A1	Obtain both results or equivalent	
		i		M1	Obtain and solve a quadratic in x^2 or y^2 or solve by inspection	
		i	$3 + 2i$ and $-3 - 2i$ or $\pm(3 + 2i)$	A1 A1	Obtain correct answers as complex numbers S.C. $\pm(3 \pm 2i)$ scores A1 Examiner's Comments Most candidates knew the method for finding the square roots of a complex number, the most common error was finding e.g. $x^2 = 9$, but then not square rooting to get the correct real part.	
		ii		M1	Solve using quadratic formula or complete square	
		ii	$(4 \pm 2\sqrt{5 + 12i})/2$	A1	Obtain correct answers, or simpler version	
		ii		M1	Use result(s) from (i)	
		ii	$5 + 2i$ and $-1 - 2i$ or $2 \pm (3 + 2i)$	A1 A1	Obtain correct answers Examiner's Comments Those candidates who completed the square to solve the quadratic usually solved correctly seeing the answers from (i) could be used. Those who tried the quadratic formula often made sign errors, so could see that "hence" was not suitable. This should have prompted them to check their working. Some gave 4 answers from using all combinations of $\pm$ signs.	If more than 2 roots A0 A0



Question			Answer/Indicative content	Marks	Part marks and guidance	
			Total	10		
48			$\sqrt{3}$ $2\sqrt{3}$ $3 - \sqrt{3}i$ $-\sqrt{3}i$	M1 A1 M1 A1FT B1FT B1FT	Use correct trig expression Obtain correct answer Correct expression for modulus Obtain correct answer a.e.f. Correct conjugate seen or implied Correct answer Examiner's Comments Most candidates found the value of a correctly, the most common errors were using $\tan^{-1}(\frac{1}{6}\pi)$ or $\frac{3}{a} = \tan(\frac{1}{6}\pi)$. Candidates usually earned some or all of the marks for the remaining requests, with the notation for the modulus and the complex conjugate understood by the majority of candidates. The most frequent error was giving the conjugate as $-3 - \sqrt{3}i$.	
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Question			Answer/Indicative content	Marks	Part marks and guidance	
49			$x^2 - y^2 = 11$ and $xy = 6\sqrt{5}$ $\pm(2\sqrt{5} + 3i)$	M1	Attempt to equate real and imaginary parts of $(x + iy)^2$ and $11 + 12\sqrt{5}$	
				A1	Obtain both results cao	
				M1*	Obtain a quadratic in x^2 or y^2	
				DM1	Solve a 3 term quadratic to obtain a value for x or y	
				A1	Obtain 1 correct answer as complex number	
				A1	Obtain only the other correct answer	
					Examiner's Comments	
					Most candidates showed knowledge of the method required to find the square roots of a complex number and made good progress with the question. The main errors were either to give only one root, or more than two roots, eg $\pm(2\sqrt{5} \pm 3i)$. A few found the correct values of x and y, but did not write their answers as complex numbers.	
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Question			Answer/Indicative content	Marks	Part marks and guidance	
50		i		M1	Use $\arg(z - a) = \theta$ in equation for l /condone missing brackets	
		i	$\arg(z - 3i) = \frac{1}{4}\pi$	A1	Obtain correct answer	
		i		M1	Use $ z - a = k$ in equation for C , k must be real	
		i	$ z - 3i = 3$	A1	Obtain correct answer	
					Examiner's Comments	
					This part proved quite testing, with many candidates not realising the form of answer that was required. Some gave cartesian forms for C and l .	
		ii	$ z - 3i \leq 3$ or e.g. $x^2 + (y - 3)^2 \leq 9$	B1	Obtain correct inequality, or answer consistent with sensible (i)	
		ii	$\frac{1}{4}\pi \leq \arg(z - 3i) \leq \frac{1}{2}\pi$ or $y \geq x + 3, x \geq 0$	B1B1	Each correct single inequality, or answer consistent with sensible (i) SC if $<$ used consistently, but otherwise all correct, B2	
					Examiner's Comments	
					Candidates who made some progress with part (i) usually scored some marks in part (ii). The most frequent error was the omission of the third inequality $\arg(z - 3i) \leq \frac{1}{2}\pi$.	
Total				7		



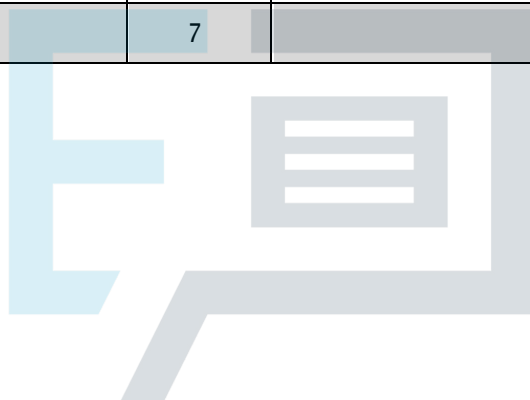
Question			Answer/Indicative content	Marks	Part marks and guidance	
51		i	$ z = \sqrt{58}$ or 7.62	B1	Obtain correct value, 3 s.f. or better	
		i	$\arg z = 23.2(^{\circ})$ or 0.405 or 0.129π	B1	Obtain correct value, 3 s.f. or better <u>Examiner's Comments</u> The modulus and argument were both found correctly by most candidates. A common incorrect value for the modulus was $\sqrt{40}$, while a significant minority who gave either or both answers as decimals did not give sufficient significant figures.	$\arctan(3/7)$ gets B0
		ii	Either	M1	Multiply numerator & denominator by conjugate	
		ii	$\frac{1}{17}(25+19i)$ or $\frac{25}{17} + \frac{19}{17}i$	A1	Obtain correct numerator or real part	$\frac{28+19i-3}{16+1}$ gets A0
		ii	Or	A1	Obtain correct denominator or imaginary part	
		ii		M1	Find and attempt to solve a pair of simultaneous equations for real and imaginary parts of answer	
		ii	$\frac{25}{17}$ and $\frac{19}{17}$	A1A1	Obtain correct answers <u>Examiner's Comments</u> Most candidates multiplied by the conjugate of the denominator, but many then gave the value of $(4+i)(4-i)$ as 15, rather than 17	
Total				5		



Question			Answer/Indicative content	Marks	Part marks and guidance	
52		i		B1	Half line,	Not line segment e.g (0, 0) to (2, 2)
		i		B1	starting at (2, 2) with +ve slope upwards	Must be half line
		i		B1	Vertical line	
		i		B1	Clearly $x = 5$ (must be vertical)	
					<u>Examiner's Comments</u> Most understood that C_1 was a line, although the complete line thorough (2, 2) or a half starting at e.g. (-2, -2) was quite common. C_2 was frequently thought to be a circle, or a horizontal straight line. Those who recognised a vertical line usually identified its intersection with the x-axis correctly, a few giving the intersection at (10, 0).	
		ii		B1	Shade below sloping line and above horizontal through their (2, 2)	Could be for a line segment, could be earned if C_2 horizontal
		ii		B1	To right of their vertical line	



Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii		B1	Completely correct diagram <u>Examiner's Comments</u> Errors in part (i) often meant that part (ii) could score few marks, while for those with a correct part (i), the shading was often incorrect. The two most common errors being to shade to the left of a vertical line or not to appreciate that the horizontal through (2, 2) was significant.	6/6 must be earned so far
			Total	7		



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